

Enterprise Network Vendor Market Analysis

Cisco as benchmark — portfolio, technology, licensing, pricing, TCO, regional fit (India · Europe · United States) and procurement guidance

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Planning exchange rates (mid-market, Sept 2026): USD 1 = INR 95 · USD 1 = EUR 0.87

Prepared from the perspective of a Senior Network Solutions Architect / Enterprise Procurement Consultant. Vendor-neutral; no vendor sponsorship.

How to read the price labels used throughout this document	
Confirmed / Public MSRP	A price published by the vendor, or a manufacturer list price (MSRP) shown by a named reseller, with the source and date.
Reseller price	An advertised price at a named authorised reseller (e.g., CDW, SHI) on the research date. Not a negotiated enterprise price.
Market estimate	The author's estimated street/partner price range derived from reseller listings, analyst commentary and typical discount bands. Use for budgeting only.
Enterprise quote	Not publicly disclosed – enterprise/partner quotation required. Final negotiated prices typically land 30–60% below list for large deals.

Important: pricing, licensing and product names change frequently. Every figure here should be re-validated with at least two authorised partners per region before a purchase decision. TCO models are illustrative, built on stated assumptions, and are not vendor quotations.

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1. Executive Summary

The enterprise networking market in 2026 has consolidated around a handful of full-stack platforms. Two 2025 transactions reshaped the competitive map against Cisco: **HPE completed its US\$14 billion acquisition of Juniper Networks on 2 July 2025**, doubling HPE's networking business and adding routing, data-center switching and firewalls to the Aruba campus/WLAN portfolio; and **Arista acquired the VeloCloud SD-WAN business from Broadcom (closed 1 July 2025)**, giving Arista a WAN/branch offer to sit beside its data-center and campus switching. Broadcom's VMware unit has therefore exited SD-WAN and is now relevant only for NSX data-center network virtualisation inside VMware Cloud Foundation.

Headline findings

Question	Short answer	Why
Strongest overall Cisco alternative	HPE Networking (Aruba + Juniper)	Only competitor with campus, WLAN, SD-WAN, SSE, routing, DC fabric and firewalls under one owner; strong AIOps (Mist, Central).
Best price/performance	Fortinet (campus/branch/security); Arista (data center)	Fortinet's ASIC-based FortiGate bundles security + SD-WAN in one licence; Arista leads DC throughput per dollar and operational simplicity.
Lowest modelled 5-year TCO (enterprise-grade)	Fortinet (Huawei similar where it can be procured)	Lowest subscription load per device and single-OS operations; see Section 13.
Strongest SD-WAN	Fortinet, Cisco (Catalyst SD-WAN/Meraki), HPE EdgeConnect, Palo Alto Prisma SD-WAN, Versa	All are long-running Gartner SD-WAN Leaders; Fortinet best for security-led branches, HPE EdgeConnect for WAN optimisation, Versa for service providers.
Strongest data-center networking	Arista; Cisco Nexus/ACI and Juniper (Apstra/QFX) close behind	EOS single image, CloudVision, 400G/800G AI fabrics, very low defect rates.
Strongest WLAN/campus	HPE (Aruba + Juniper Mist), Cisco	AI-driven assurance, Wi-Fi 7 portfolios, mature NAC (ClearPass / ISE).
Strongest security ecosystem	Palo Alto Networks; Fortinet; Cisco (with Splunk)	Threat-prevention depth (PAN), integrated fabric (Fortinet), SOC/XDR breadth (Cisco + Splunk).
Best automation/AIOps	Juniper Mist (Marvis) and Arista CloudVision	Most mature AI-assisted troubleshooting and state-streaming automation.
Strongest in India	Cisco (installed base), then HPE Aruba, Fortinet	Cisco dominates BFSI/government/IT-ITES; Fortinet and Aruba gain on cost; Huawei restricted in telecom.
Strongest in Europe	Cisco, HPE (Aruba + Juniper), Fortinet; Nokia/Ericsson in carriers	Broad partner coverage; Huawei restricted in many EU 5G/critical networks.
Strongest in the US	Cisco; Arista in DC; Palo Alto in security	Federal certifications, largest partner base, hyperscaler adoption of Arista.
Best for global enterprise	Cisco or HPE Networking	Global depot/RMA, local TAC, consistent licensing worldwide.
Best for telecom	Nokia, Juniper (HPE), Cisco; Ciena for optical	SR-MPLS/SRv6 routing scale, IP/optical convergence; Huawei restricted by India 'trusted source' and EU 5G Toolbox rules.
Best for cost-sensitive organisations	Fortinet (security-led); Ubiquiti/MikroTik for SMB only	Lowest acquisition cost; accept reduced support depth below enterprise grade.
New enterprise network today	Single-vendor campus/branch + best-of-breed DC + converged SASE	Typical answer: HPE or Cisco for campus/WLAN, Arista for DC, Fortinet or Palo Alto for security/SASE — validated per region.

Modelled 5-year TCO snapshot (Scenario B – 500-user campus, USD, US pricing, excl. tax)

Fortinet ≈ \$0.52M · Huawei* ≈ \$0.53M · Extreme + FortiGate ≈ \$0.59M · HPE ≈ \$0.65M · Arista + partner NGFW ≈ \$0.67M · Cisco ≈ \$0.76M. Cisco carries the highest TCO in every scenario, typically 25–45% above the lowest enterprise-grade option (Fortinet) and ~15–18% above HPE, but also the lowest migration risk for Cisco-skilled teams. *Huawei is shown for reference only; it is restricted or politically difficult in the US, much of Europe and Indian telecom networks.

Scope, method and limitations

- Sources prioritised: vendor documentation and datasheets, vendor/partner price pages, authorised resellers (CDW, SHI, Ubiquiti store), Gartner Magic Quadrant announcements (2024–2025), and reputable trade press. Sources and dates are listed in the Appendix.
- Most enterprise vendors (Cisco Catalyst, Juniper, Arista, Huawei, Nokia, Extreme) do not publish full price lists; where so, figures are labelled **Market estimate** or **Enterprise quote**.
- TCO uses a transparent bottom-up model: hardware, optics, spares, installation/professional services, training, licences/subscriptions, support, security services, NAC, SASE, management platform and operations labour, with 4% annual subscription escalation from Year 2.
- Scores (1–10) are the author's analytical judgement informed by the research; they should be recalibrated with the organisation's own weights.

2. Vendor Landscape

2.1 Vendors evaluated and why

Vendor	Ownership / status (Oct 2026)	Core focus vs. Cisco	Primary categories
Cisco	Independent; owns Meraki, Splunk	Benchmark: broadest portfolio	LAN, campus, DC, WAN, SD-WAN, SASE, WLAN, security, routing, SP, automation, NMS
HPE Aruba Networking	HPE Networking segment	Campus/WLAN/NAC/SD-WAN/SSE	LAN, campus, WLAN, SD-WAN (EdgeConnect), SASE/SSE, NAC, cloud NMS
Juniper Networks	Part of HPE since 2 Jul 2025 (Rami Rahim leads HPE Networking)	Routing, DC, Mist AI campus, SRX	Routing, SP, DC (QFX/Apstra), campus (EX/Mist), security (SRX), AIOps
Arista Networks	Independent; acquired VeloCloud (Jul 2025)	DC & AI fabrics, campus, now SD-WAN	DC, campus, WLAN, SD-WAN, NDR, automation
Huawei	Independent (China)	Full stack, aggressive pricing	LAN, campus, DC, WLAN, routing, SP, security, SD-WAN
Nokia	Independent (Finland); owns Infinera (2025)	SP IP routing, optical, DC fabric (SR Linux)	SP/carrier, routing, DC, private wireless
Extreme Networks	Independent	Campus/WLAN, fabric, cloud mgmt	LAN, campus, WLAN, fabric, NAC, cloud NMS
Fortinet	Independent	Security-led networking (one OS)	NGFW, SD-WAN, SASE, LAN (FortiSwitch), WLAN (FortiAP), NAC
Palo Alto Networks	Independent	Security platform, Prisma SASE	NGFW, SASE/SSE, SD-WAN, ZTNA
Check Point	Independent	Security gateways, Harmony SASE	NGFW, SSE, management
Versa Networks	Private	Unified SASE/SD-WAN, SP-friendly	SD-WAN, SASE/SSE, branch
VMware by Broadcom	Broadcom; SD-WAN divested to Arista	NSX network virtualisation (inside VCF)	DC network virtualisation, micro-segmentation
Ciena	Independent	Optical transport, routing/access for SPs	SP/carrier, optical, metro
Dell Technologies	Independent	Open networking (Enterprise SONiC, OS10)	DC switching, campus (limited)
Ubiquiti	Independent	Prosumer/SMB, no subscriptions	SMB LAN, WLAN, gateways
Ruckus Networks	Part of CommScope restructuring	WLAN for hospitality/education/MDU	WLAN, campus switching (ICX)
MikroTik	Private (Latvia)	Low-cost routing/switching	SMB/ISP routing, switching
Cato Networks (added)	Private	Cloud-native single-vendor SASE	SASE/SSE, SD-WAN — added: Gartner SASE Leader 2025
Zscaler (added)	Independent	SSE (ZIA/ZPA), zero trust	SSE/ZTNA — added: most common SSE paired with Cisco/Arista networks

Netskope is also a 2025 Gartner SASE Platforms Leader and is a credible SSE alternative; Ericsson and ZTE matter only in carrier RAN contexts and are out of scope.

2.2 Category map

Category	Leaders / primary contenders	Challengers / niche
Enterprise LAN / Campus	Cisco, HPE Aruba, Juniper Mist, Huawei	Extreme, Arista, Fortinet, Ruckus, Dell
Wi-Fi / WLAN	Cisco (Catalyst/Meraki), HPE Aruba, Juniper Mist, Huawei	Extreme, Fortinet, Ruckus, Arista, Ubiquiti (SMB)
Data Center Networking	Arista, Cisco (Nexus/ACI/NDFC), Juniper (QFX/Apstra)	Nokia (SR Linux), Huawei (CloudEngine), Dell (SONiC), Extreme
WAN / Routing	Cisco, Juniper, Nokia, Huawei	Arista (7280R), Ciena (access/metro), MikroTik (SMB/ISP)
SD-WAN	Fortinet, Cisco, HPE (EdgeConnect), Palo Alto, Versa, Arista VeloCloud	Huawei, Extreme, Check Point, Cato

Category	Leaders / primary contenders	Challengers / niche
SASE / SSE	Palo Alto (Prisma), Fortinet, Cato, Netskope, Zscaler (SSE), Cisco (Secure Access)	HPE (Aruba SSE), Versa, Check Point (Harmony)
Network Security (NGFW)	Palo Alto, Fortinet, Check Point, Cisco	Juniper SRX, Huawei USG, Sophos (SMB)
Network Automation	Arista CloudVision, Juniper Apstra, Cisco NDFC/Catalyst Center	Nokia EDA, Extreme, Ansible/Terraform ecosystems
Network Management / AIOps	Juniper Mist/Marvis, HPE Aruba Central, Cisco Meraki/Catalyst Center + ThousandEyes	Extreme Platform ONE, Huawei iMaster NCE, Fortinet FortiManager
Service Provider / Carrier	Nokia, Cisco, Juniper, Ciena, Huawei (where allowed)	Arista (cloud/SP peering), Ericsson (RAN-adjacent)

Analyst signals (2025): Gartner's 2025 Magic Quadrant for Enterprise Wired and Wireless LAN Infrastructure (25 June 2025) named Huawei a Leader for a third year as the only non-North American Leader; listings also show HPE (Aruba), Juniper and Fortinet as Leaders. The 2024 SD-WAN Magic Quadrant Leaders were Broadcom (VeloCloud), Cisco, Fortinet, HPE, Palo Alto Networks and Versa. The 9 July 2025 SASE Platforms Magic Quadrant Leaders announced by vendors include Palo Alto Networks (third consecutive year), Fortinet, Cato Networks and Netskope.

3. Product Comparison

Major product families per vendor. A dash (–) means the vendor has no first-party offer in that area.

3.1 Switching and routing

Vendor	Access / PoE	Distribution / Core / Modular	Data center	Enterprise / branch / edge routing	SP routing
Cisco	Catalyst 9200/9300/9350; Meraki MS130/150/390	Catalyst 9400/9500/9600 (modular 9400/9600)	Nexus 9000 (ACI/NX-OS), Nexus HyperFabric, 8000 series	Catalyst 8000 edge (8200/8300/8500), Meraki MX	ASR 9000, Cisco 8000, NCS 540/5500/5700
HPE Aruba	CX 6000/6100/6200/6300	CX 6400 (modular), CX 8xxx, CX 9300	CX 10000 (DPU/AMD Pensando), CX 8360/9300	EdgeConnect, 9000-series gateways	– (via Juniper)
Juniper (HPE)	EX2300/EX4100/EX4400	EX4650, EX9200 (modular)	QFX5120/5130/5240/5700, PTX for DC/AI; Apstra	SRX branch, Session Smart Router (SSR)	MX, PTX, ACX
Arista	CCS-710/720XP/722 (campus)	750 series (modular), 7050X	7050X/7060X/7280R/7800R, 7700R4 AI spine	VeloCloud Edge; 7280R edge	7280R/7800R (peering/edge)
Huawei	S5735/S5736/S6730	S12700E, CloudEngine S16700	CloudEngine CE6800/CE8800/CE16800	NetEngine AR6000/AR8000	NetEngine 8000, NE40E
Nokia	– (limited)	7250 IXR	7220/7250 IXR with SR Linux, EDA	7705 SAR, 7750 SR-a	7750 SR, FP5-based 7750/7950
Extreme	5320/5420/5520/4120/4220	7520/7720, X870	8000 series, SLX 9000	XA1400, Extreme SD-WAN	SLX (limited)
Fortinet	FortiSwitch 100/200/400 series (PoE)	FortiSwitch 500/1000/2000	FortiSwitch 3000 (limited DC)	FortiGate (routing + SD-WAN)	–
Palo Alto / Check Point / Versa	–	–	–	PA/Quantum gateways; Versa CSG appliances route	Versa (SP-hosted SD-WAN)
Dell	PowerSwitch N-series	S/Z-series	Z9864F (800G), S5200; Enterprise SONiC / OS10	–	–
Ubiquiti	UniFi Pro Max, Pro XG, Enterprise	UniFi Aggregation / ECS	–	UniFi Gateways (UDM, UXG)	– (UISP for WISPs)
Ruckus	ICX 7150/7550/8200	ICX 7650/7850	–	–	–
MikroTik	CRS3xx/CRS5xx	CRS5xx (100G)	–	hEX, RB5009, CCR2004/2116/2216	CCR (small ISP)
Ciena	–	–	–	–	8100/8700 routers, 3900/5000 access, WaveLogic optical

3.2 Wireless, SD-WAN and security

Vendor	Wi-Fi APs (6/6E/7)	WLAN control / cloud	SD-WAN	Firewall / NGFW	SSE / SASE, ZTNA, SWG, CASB, DLP
Cisco	Catalyst/Meraki CW9172/9176/9178 (Wi-Fi 7), outdoor CW9163E	Catalyst 9800 WLC, Meraki dashboard	Catalyst SD-WAN (Viptela), Meraki SD-WAN	Secure Firewall (FTD) 1200/3100/4200, Meraki MX	Cisco Secure Access (SSE), Umbrella, Duo; ISE NAC
HPE Aruba	AP-500/600/700 series incl. Wi-Fi 7; outdoor AP-570/580	Central (cloud), Mobility Conductor	EdgeConnect SD-WAN (Silver Peak)	Gateways with IDS/IPS; Juniper SRX	HPE Aruba SSE (Axis); ClearPass NAC
Juniper (HPE)	Mist AP24/34/45/47 (Wi-Fi 7)	Mist cloud, Marvis AI	Session Smart (SSR) SD-WAN	SRX 300–5000, vSRX	Secure Edge (SSE); Mist Access Assurance (NAC)
Arista	C-360/C-460 Wi-Fi 6E/7	CV-CUE	VeloCloud SD-WAN	– (partner)	AGNI (NAC); SSE via partners
Huawei	AirEngine Wi-Fi 7 (incl. 5-radio AP)	iMaster NCE-Campus, AC6800	HiSecEngine/AR SD-WAN	HiSecEngine USG6000/12000	Limited SASE outside China
Extreme	AP3000/4000/5000 (Wi-Fi 6E/7)	ExtremeCloud IQ / Platform ONE	Extreme SD-WAN (Ipanema)	– (partner)	ExtremeControl NAC, Universal ZTNA
Fortinet	FortiAP 231G/441K (Wi-Fi 7)	FortiGate controller, FortiLAN Cloud	FortiGate Secure SD-WAN	FortiGate 40F–7000F	FortiSASE, ZTNA, SWG, CASB, DLP; FortiNAC
Palo Alto	–	–	Prisma SD-WAN (CloudGenix)	PA-400/1400/3400/5400/7500, VM/CN-series	Prisma Access/SASE, ZTNA 2.0, CASB, DLP

Vendor	Wi-Fi APs (6/6E/7)	WLAN control / cloud	SD-WAN	Firewall / NGFW	SSE / SASE, ZTNA, SWG, CASB, DLP
Check Point	–	–	Quantum SD-WAN	Quantum Force gateways	Harmony SASE (Perimeter 81)
Versa	Versa WLAN (limited)	Versa Director	Versa Secure SD-WAN	Versa NGFW	Versa Unified SASE/SSE
Ruckus	R770/R670 Wi-Fi 7, T-series outdoor	RUCKUS One, SmartZone	–	–	–
Ubiquiti	U7 Pro/Pro Max, outdoor U7 Outdoor	UniFi Network (self-hosted/cloud)	Basic Site Magic VPN	Basic UniFi gateway IPS	–
MikroTik	cAP/wAP (Wi-Fi 6)	CAPsMAN	–	RouterOS firewall (stateful)	–
Cato / Zscaler	–	–	Cato Socket (SD-WAN); Zscaler Zero Trust Branch	Cloud FWaaS	Full SASE (Cato); SSE leader (Zscaler)

3.3 Data center, management and automation

Vendor	Spine-leaf / EVPN-VXLAN	Virtualisation / load balancing	NMS / AIOps	APIs & automation (Terraform, Ansible, Python, NETCONF, REST, gNMI)
Cisco	Yes – ACI or NX-OS EVPN with NDFC	ACI multi-site; LB via partners (F5)	Catalyst Center, Meraki, Nexus Dashboard, ThousandEyes, Splunk	All: Terraform providers, Ansible, pyATS/Genie, NETCONF/RESTCONF, gNMI
HPE Aruba / Juniper	Yes – CX EVPN; Juniper Apstra (multi-vendor intent)	CX 10000 distributed services; LB via partners	Central, Mist/Marvis, Apstra, Paragon	All; Apstra and Mist APIs are strong; PyEZ, NETCONF native on Junos
Arista	Yes – EOS EVPN/VXLAN, AVD	Partner LB	CloudVision (CVaaS), NetDL	All; eAPI, gNMI/OpenConfig native, AVD (Ansible) reference
Huawei	Yes – CloudEngine EVPN	iMaster NCE-Fabric	iMaster NCE, CampusInsight	NETCONF/RESTCONF, Ansible; smaller community
Nokia	Yes – SR Linux + EDA (Kubernetes-based)	–	EDA, NSP	Model-driven, gNMI-first; strong for NetDevOps teams
Extreme	Yes – fabric (SPB) and EVPN	–	Platform ONE / ExtremeCloud IQ	REST, Ansible, Python; fewer Terraform resources
Fortinet	Limited DC fabric	FortiADC (LB)	FortiManager/Analyzer, FortiAIOps	REST, Ansible, Terraform (FortiOS)
Dell	Yes – SONiC EVPN	–	SmartFabric	SONiC open APIs, gNMI
VMware by Broadcom	Overlay (Geneve) on any underlay	NSX routing, Avi LB (separate)	VCF Operations	Terraform, PowerCLI, REST

4. Technical Feature Matrix

Legend: **N** = Native (in base OS) · **L** = Licensed tier/subscription · **A** = Add-on product · **3P** = Third-party/partner · **NA** = Not available. Values reflect flagship enterprise platforms; individual models vary.

Capability	Cisco	Aruba	Juniper	Arista	Huawei	Fortinet	Extreme	Nokia	Dell	Ubiquiti
Max switching capacity (flagship)	51.2T+ (Nexus/8000)	~25.6T	~51.2T (QFX/PTX)	51.2T (7060X6/7800R4)	~51.2T	~6.4T	~12.8T	51.2T+	51.2T (Z9864F)	<1T
Port speeds	1G–800G	1G–400G	1G–800G	1G–800G	1G–800G	1G–100G	1G–400G	10G–800G	10G–800G	1G–25G
PoE / PoE+ / PoE++ (90W)	N	N	N	N	N	N	N	NA	Limited	N
VLAN, STP/RSTP/MSTP	N	N	N	N	N	N	N	N	N	N
VRF	N/L	N	N	N	N	N	N	N	N	Limited
OSPF / BGP	N (BGP on Adv. on some access)	N	N	N	N	N	L (Core)	N	N	Basic
IS-IS	L	N (CX)	N	N	N	N	L	N	N	NA
EIGRP	N (Cisco only)	NA	NA	NA	NA	NA	NA	NA	NA	NA
MPLS / Segment Routing	L	Limited	N	N	N	Limited	Limited	N	Limited	NA
EVPN / VXLAN	N/L	N	N	N	N	N	N	N	N	NA
QoS, Multicast, IPv6	N	N	N	N	N	N	N	N	N	Basic
SDN / intent fabric	L (SD-Access, ACI)	L	L (Apstra)	L (CloudVision)	L	N (Fabric)	N (SPB)	L (EDA)	A	NA
SD-WAN	L	L	L	L (VeloCloud)	L	N (in FortiOS)	L	NA	NA	Basic
SASE / SSE	L	L	L	3P	Limited	L	3P	NA	NA	NA
Zero trust / segmentation	L (SD-Access, ISE)	L (ClearPass, dynamic seg.)	L	L (AGNI, MSS)	L	L	N (fabric) / L	N	Limited	NA
NAC	A (ISE)	A (ClearPass)	L (Mist Access Assurance)	L (AGNI)	A	A (FortiNAC)	A (ExtremeControl)	NA	NA	NA
Streaming telemetry (gNMI)	N	N	N	N	N	Limited	N	N	N	NA
AI / AIOps	L	L	L (Marvis)	L (AVA)	L	L	L	L	Limited	Basic
Cloud mgmt / multi-cloud	L	L	L	L	L	L	L	L	Limited	N
APIs (REST/NETCONF)	N	N	N	N	N	N	N	N	N	Limited
Stacking / virtual chassis	N (StackWise)	N (VSF/VSX)	N (VC)	N (MLAG/stack)	N (iStack/CSS)	N	N	N (MC-LAG)	N (VLT)	Limited
HA / redundancy (PSU, ISSU)	N	N	N	N	N	N	N	N	N	Basic
MACsec / IPsec encryption	N/L	N	N	N	N	N	N	N	N	IPsec only
Integrated NGFW/IPS	A	L (gateway)	A (SRX)	3P	A	N/L	3P	NA	NA	Basic

Switching capacity reflects top-end data-center/SP chassis or 51.2T single-chip systems announced or shipping by 2025–2026; campus figures are far lower. Verify per-SKU datasheets. Palo Alto, Check Point, Versa, Cato and Zscaler are security/SASE vendors and are compared in Sections 3.2 and 5.3.

5. Licensing Comparison

Rule for every evaluation: **Hardware cost ≠ Software licence ≠ Cloud management ≠ Support ≠ Security subscription**. Each is a separate line item, has a separate term, and is renewed (or not) independently. Many vendors bundle them on a single SKU for convenience, which hides the split.

5.1 Licensing models at a glance

Model	Cisco	HPE Aruba	Juniper	Arista	Huawei	Fortinet	Extreme	Palo Alto	Ubiquiti / MikroTik
Perpetual OS licence	Yes (Network Ess./Adv.)	Yes (AOS-CX base)	Yes (Junos base)	Yes (EOS)	Yes (base + some feature)	Yes (FortiOS)	Yes (legacy) / Platform ONE	Yes (PAN-OS base)	Yes, no subs
Term subscription	Mandatory (Catalyst/DNA 3/5/7 yr)	Central Foundation/Advanced	Mist subs; Flex tiers	CloudVision, AGNI	Optional (NCE, SD-WAN)	FortiGuard bundles	Platform ONE tiers	Mandatory for security value	No
Device-based	Yes	Yes	Yes	Yes	Yes	Yes	Yes (pooled)	Yes	N/A
Port-based	Some (port tiers)	By port count tier	By model tier	Tiered	Some	No	Tiered by device class	No	No
User-based	SASE/Duo/ISE endpoint	SSE per user, ClearPass per endpoint	Secure Edge per user	AGNI per user	Limited	FortiSASE per user	ZTNA per user	Prisma Access per user	No
Bandwidth-based	SD-WAN tiers by throughput	EdgeConnect by bandwidth	SSR by bandwidth	VeloCloud by bandwidth	SD-WAN	No (appliance)	SD-WAN	Prisma SD-WAN / remote network	No
Cloud mgmt licence	Meraki (mandatory)	Central (mandatory for cloud)	Mist (mandatory for APs)	CVaaS	NCE cloud	FortiCloud/ FortiLAN	ExtremeCloud IQ	Strata Cloud Manager	Free
Security subscription	Secure Firewall Threat/Malware/URL	Gateway IDS/IPS	SRX ATP	Partner	USG bundles	UTP/Enterprise bundles	Partner	Core Security bundle	Basic IPS (Ubiquiti)
Support contract	Smart Net Total Care / Solution Support	Foundation Care	Juniper Care	A-Care	Huawei Co-Care	FortiCare Premium/Elite	ExtremeWorks	Standard/Premium	Warranty only (+ optional UI Care)

5.2 Cisco licensing in detail

Element	How it works	Procurement implication
Network Stack (Network Essentials / Network Advantage)	Perpetual licence tied to Catalyst 9000 hardware; Essentials covers L2/basic L3, Advantage adds full routing, VRF, MPLS/EVPN, etc.	Choose tier up front; upgrading later costs more than buying Advantage at purchase for distribution/core.
Cisco Catalyst / DNA subscription (Essentials / Advantage)	Per Cisco ordering guides, new Catalyst 9000 orders require a matching-tier term subscription of 3, 5 or 7 years alongside the perpetual stack licence.	This is the 'hidden' recurring cost. Renewal pricing is the largest Cisco campus OPEX lever.
Catalyst Center (formerly DNA Center)	Renamed in 2024. Meters per managed device; tiers Essentials, Advantage, Premier.	Appliance/VM capex plus device subscriptions; size for growth.
Meraki licensing	Every Meraki device needs an active licence for dashboard management and support (co-term or per-device). MX licence tiers: Enterprise and Advanced Security (adds IDS/IPS, AMP, content filtering).	If the licence lapses, devices stop passing traffic after the grace period — treat as mandatory OPEX.
Cisco Networking Subscription (unified)	Newer offer (CISCO-NETWORK-SUB) letting Meraki and Catalyst devices use unified Essentials/Advantage licences (e.g., Wi-Fi 7 APs, smart switching), managed in cloud or on-prem.	Simplifies hybrid estates; evaluate vs. legacy SKUs and ELA.
Security subscriptions	Secure Firewall threat/malware/URL, Umbrella/Secure Access per user, Duo per user, ISE Essentials/Advantage/Premier per endpoint.	ISE and SSE are separate budgets from LAN; often 20–35% of total Cisco spend.
Smart Licensing / Smart Licensing Using Policy	Licences live in Smart Accounts/Virtual Accounts; SLUP makes many tiers unenforced but reportable.	Compliance is audited at renewal/true-up; keep Smart Account hygiene.

Element	How it works	Procurement implication
Support (Smart Net Total Care / Solution Support)	8x5xNBD to 24x7x4 hardware replacement + TAC + software updates.	Typically priced as an annual % of list; renew in sync (co-term).
Enterprise Agreement (EA)	Multi-year commitment across networking, security, Splunk etc.	Better unit pricing but 'true-forward' penalises shrinking device counts.

5.3 Competitor licensing equivalents

Vendor	Licensing model	What is mandatory vs. optional	Watch-outs
HPE Aruba Networking	Perpetual AOS-CX/AOS-10 + Central subscription (Foundation / Advanced) per device, 1/3/5/7/10 yrs; ClearPass perpetual or subscription per endpoint; EdgeConnect by bandwidth tier; SSE per user.	AOS-10 APs require Central; switches can run without cloud.	Central Advanced is meaningfully dearer than Foundation; check gateway licences.
Juniper (HPE)	Junos perpetual base + Flex Standard/Advanced/Premium (term) for features; Mist Wi-Fi/Wired Assurance per device; Marvis add-on; Apstra per device; SRX security subs.	Mist Wi-Fi Assurance is mandatory for Mist APs; Marvis needs Wired Assurance.	Roadmap convergence with Aruba Central may change SKUs — insist on price protection.
Arista	EOS perpetual with feature licences (e.g., routing scale); CloudVision as-a-Service per device; AGNI per user; VeloCloud per edge/bandwidth.	CloudVision optional but expected.	Fewer bundles; optics pricing matters.
Huawei	Perpetual hardware + feature licences; iMaster NCE per device; optional SnS (Software Subscription & Support).	Mostly perpetual.	SnS needed for upgrades; regulatory constraints outweigh price.
Fortinet	Perpetual FortiOS; FortiGuard bundles (ATP, UTP, Enterprise) 1/3/5 yrs; FortiSwitch/FortiAP mostly licence-free when managed by FortiGate; FortiSASE per user.	Bundles are optional but needed for security value.	Bundled SKUs (e.g., FG-60F-BDL-950-36) are cheaper than à la carte.
Extreme Networks	Extreme Platform ONE Networking: Standard, Advanced, Premium; Standard required for ExtremeCloud IQ management; licences pooled and portable across devices; legacy perpetual feature licences still supported on non-Universal switches.	Standard tier needed for cloud management.	Pooled licences reduce stranded cost.
Palo Alto Networks	Appliance + per-device subscriptions (Advanced Threat Prevention, Advanced URL, Advanced WildFire, DNS, SD-WAN) — bundled as Core Security; Prisma Access per user/site.	Without subscriptions the device is a basic stateful firewall.	Over 5 years subscriptions can be 3–4× the appliance; negotiate renewal caps.
Check Point	Appliance + Quantum security packages (NGFW/NGTP/SandBlast); Harmony SASE per user.	Packages required for threat prevention.	Management server licence separate.
Versa	Subscription per site/bandwidth (Secure SD-WAN, SASE tiers); often delivered via service providers.	Subscription mandatory.	Pricing often bundled in managed service.
VMware by Broadcom	Subscription only; per physical core (16-core minimum per CPU). NSX networking ships inside VMware Cloud Foundation; advanced security (vDefend) is an additional per-core attach.	Must buy VCF to get NSX.	Major cost increases for many customers since 2024.
Ubiquiti / MikroTik	No recurring licences; RouterOS licence level bundled with hardware (MikroTik).	None.	No enterprise TAC; optional Ubiquiti UI Care (e.g., US\$259 per Pro Max 48 PoE for 5 yrs).
Cato / Zscaler	Pure SaaS per user/site/bandwidth.	Subscription mandatory.	Cloud dependency; data-residency (PoP) checks.

6. Hardware Pricing

Representative, comparable SKUs. List/MSRP and reseller prices are from public listings on the dates shown; **all other figures are estimates** for budgeting.

Vendor / model	Configuration	Throughput / capacity	Price (USD)	Price type	Source (date)
Cisco C9300-48P-E	48×1G PoE+ (437W default PSU), modular uplinks, Network Essentials	StackWise-480, 256 Gbps switching	\$2,765 (was \$2,925)	Reseller price (hardware only; DNA/Catalyst sub extra)	IT Bargain Center listing, viewed 8 Oct 2026
Cisco C9300-48P-E	Same, UK	—	£5,065 ex VAT (special) / £5,331 regular	Reseller price (UK)	Sync (wearesync.co.uk), viewed 8 Oct 2026
Cisco C9300-48P-E incl. DNA 3Y	Same + 3-yr DNA subscription	—	€4,350 incl. VAT	Reseller price (DE marketplace)	eBay.de Comstex, updated Aug 2025
Cisco Meraki MX67	Branch SD-WAN security appliance	~450 Mbps stateful FW	Not publicly disclosed – partner quote; est. \$600–\$900	Market estimate	—
Fortinet FortiGate 60F + 3 yr UTP bundle	10×GE, SPU; 3-yr 24×7 FortiCare + FortiGuard UTP (FG-60F-BDL-950-36)	700 Mbps threat protection	\$1,847–\$1,939 (MSRP \$2,389–\$2,508)	Reseller price + MSRP	SHI, viewed 8 Oct 2026
Palo Alto PA-440	8×GE desktop NGFW	3.0 Gbps FW / 2.2 Gbps threat	€2,490 ex VAT (≈ \$2,860)	Reseller price (FR, appliance only)	Optinoc product page, viewed 8 Oct 2026
Ubiquiti USW-Pro-Max-48-PoE	16×2.5G + 32×1G, 720W PoE++, 4×10G SFP+	L3, Etherlighting	\$1,299 base; \$1,401 incl. tariff surcharge	Vendor store price (Confirmed)	store.ui.com (US), viewed 8 Oct 2026
Ubiquiti USW-Pro-Max-48-PoE	Same, via reseller	—	\$1,516–\$1,552	Reseller price	SHI, viewed 8 Oct 2026
HPE Aruba CX 6300M 48-port PoE	48×1G Class 4 PoE, 4×50G SFP56	~880 Gbps	Not publicly disclosed; est. \$4,000–\$6,500 street	Market estimate	—
Juniper EX4100-48P	48×1G PoE+, 4×25G	~496 Gbps	Not publicly disclosed; est. \$3,500–\$6,000 street	Market estimate	—
Arista CCS-720XP-48ZC2	48×mGig PoE, 6×100G	~1.6 Tbps	Not publicly disclosed; est. \$4,000–\$7,000 street	Market estimate	—
Huawei S5735-L48P4X	48×1G PoE+, 4×10G	~176 Gbps	Not publicly disclosed; est. \$1,200–\$2,500 street	Market estimate	—
Extreme 5420M-48W-4YE	48×1G PoE+, 4×25G	~400 Gbps	Not publicly disclosed; est. \$2,500–\$4,500 street	Market estimate	—
Wi-Fi 7 enterprise AP (Cisco CW9176I / Aruba 735 / Mist AP47 / Arista C-460)	4×4 tri-band, 10GbE	—	Not publicly disclosed; est. \$900–\$1,600 street	Market estimate	—
400G DC spine (Arista 7060X5/X6, Cisco N9K-9364D, Juniper QFX5240)	32–64×400G/800G	12.8–51.2 Tbps	Enterprise quote; est. \$35,000–\$90,000 street	Market estimate	—

6.1 Additional hardware costs that are commonly missed

Item	Typical cost position	Notes
Optics (SFP/SFP+/SFP28/QSFP28/QSFP-DD)	Vendor-branded optics 3–10× third-party; 10G SR \$20–\$150 (3P vs. branded), 100G \$150–\$1,500, 400G \$600–\$4,000	Arista/Juniper/Dell tolerate compatible optics more readily; Cisco/HPE may restrict TAC support.
DAC/AOC cables	\$13–\$40 (10G DAC, Ubiquiti list) up to \$300+ for 400G	Use DAC inside racks to save optics cost.
Redundant PSUs, fans, stacking cables	+8–20% per access switch	Catalyst 9300 ships with one PSU by default.
Controllers / management appliances	\$15k–\$120k (Catalyst Center, ClearPass, FortiManager, NCE)	Cloud mgmt converts this to OPEX.
Spares pool	3–5% of hardware capex	Critical for India/remote sites where RMA can be 2–5 days.

Item	Typical cost position	Notes
Rack accessories, PDUs, UPS	1–3% of capex	—
Logistics & import duties	India: BCD 10–20% + SWS + 18% IGST (input credit available to GST-registered buyers); EU: 0% duty on most IT (ITA) + VAT 17–27%	Duty classification disputes common in India (see Section 9).

7. Software Pricing

Licence / subscription	Term	Price (USD)	Price type	Annualised	Source (date)
Meraki MX67W Advanced Security + Enterprise support (LIC-MX67W-SEC-1YR)	1 yr	\$523.99 (list \$735.57)	Reseller price	\$524	CDW, viewed 8 Oct 2026
Meraki MX67 Advanced Security (LIC-MX67-SEC-1YR)	1 yr	\$359.94 (list \$686.39)	Reseller price	\$360	Hummingbird Networks, viewed 8 Oct 2026
HPE Aruba Central Advanced, 1 switch (JZ535AAE)	1 yr	\$665 (MSRP \$825)	Reseller + MSRP	\$665	SHI, viewed 8 Oct 2026
HPE Aruba Central Advanced, 1 switch 8–12 ports (JZ526AAE)	3 yr	\$766 (MSRP \$950)	Reseller + MSRP	\$255	SHI, viewed 8 Oct 2026
HPE Networking Central Advanced, 1 switch (6400/5400 class, S0T70AAE)	5 yr	\$4,977 (list \$7,425)	Reseller price	\$995	CDW, viewed 8 Oct 2026
HPE Aruba Central Advanced, 1 WLAN gateway (S0V01AAE)	1 yr	\$2,986 (MSRP \$3,706)	Reseller + MSRP	\$2,986	SHI, viewed 8 Oct 2026
Juniper Mist Wired Assurance + Care Core, EX 48-port (SUB-EX48-1S-5Y-COR)	5 yr	\$1,085 (list \$1,363)	Reseller price	\$217	CDW, viewed 8 Oct 2026
Juniper Mist Wired Assurance + Same Day support, EX 48-port	5 yr	\$10,447 (list \$12,074)	Reseller price	\$2,089	CDW, viewed 8 Oct 2026
Juniper Mist Wi-Fi Assurance (per AP)	per month	~\$10–\$20 per AP/month	Market estimate (aggregator)	\$120–\$240	Costbench, Jun 2026
Fortinet FortiGate 60F UTP bundle (HW + 3 yr)	3 yr	\$1,847–\$1,939	Reseller price	≈ \$500 sub/yr (est.)	SHI, viewed 8 Oct 2026
Palo Alto PA-440 'Bundle Complete' (threat, URL, WildFire, GlobalProtect, SD-WAN)	1 yr	€2,890 ex VAT (≈ \$3,320)	Reseller price (FR)	≈ \$3,320	Optinoc, viewed 8 Oct 2026
Palo Alto PA-440 3-yr TCO with bundle	3 yr	€11,160 ex VAT (≈ \$12,830)	Reseller-computed	—	Optinoc, viewed 8 Oct 2026
VMware Cloud Foundation (includes NSX)	per core / yr	~\$350 per core/yr list; 16-core minimum per CPU	Market estimate (licensing advisory)	—	Redress Compliance, 2026
VMware vDefend (NSX distributed firewall) attach	per core / yr	~\$120–\$200 per core/yr on top of VCF	Market estimate	—	Redress Compliance, 2026
Cisco Catalyst/DNA Advantage (per 48-port switch)	5 yr	Not publicly disclosed – partner quote; est. \$1,500–\$3,500 street	Market estimate	\$300–\$700	—
Cisco ISE / Aruba ClearPass / FortiNAC per endpoint	per yr	Enterprise quote; est. \$2–\$15 per endpoint/yr	Market estimate	—	—
SASE / SSE per user (Prisma Access, FortiSASE, Cisco Secure Access, Zscaler, Cato, HPE SSE)	per yr	Enterprise quote; est. \$60–\$180 per user/yr depending on features and volume	Market estimate	—	—

8. Support & Maintenance

Vendor	Standard	Premium / 24x7	Software updates	Hardware replacement	Typical annual cost (% of HW list, est.)	TAC quality notes
Cisco	SNTC 8x5xNBD	24x7x4, Solution Support, Success Tracks	Requires active contract	NBD / 4-hr depots globally	10–20%	Deepest global TAC; large knowledge base.
HPE Aruba	Foundation Care NBD	24x7 / 4-hr, Aruba TAC	Included with Central / FC	NBD / 4-hr; limited lifetime warranty on many CX/Instant On	8–15%	Good; lifetime warranty reduces HW cost.
Juniper (HPE)	Juniper Care Core	Same Day / 24x7x4	With Care / subs	NBD / same-day	8–15%	Strong SP-grade JTAC; Mist bundles support into subs.
Arista	A-Care NBD	4-hr onsite	With A-Care	NBD / 4-hr	10–15%	Very well regarded; engineer-to-engineer.
Huawei	Co-Care basic	24x7 premium	SnS	NBD / 4-hr (country dependent)	5–10%	Strong in Asia/MEA; constrained in restricted markets.
Fortinet	FortiCare Premium (24x7)	FortiCare Elite (15-min SLA)	Included in bundles	NBD advance replacement	Included in bundle (~15–25% of HW/yr incl. FortiGuard)	Mixed reviews at scale; improving.
Extreme	ExtremeWorks NBD	24x7, ResidentEngineer	Included	NBD / 4-hr	8–14%	Frequently praised for in-house TAC.
Palo Alto	Standard	Premium (24x7), Focused Services	Requires support	NBD / 4-hr	15–20% (+ security subs)	Strong security TAC.
Ubiquiti	Warranty (1–2 yr)	UI Care (5-yr advance replacement, e.g., \$259/unit for Pro Max 48 PoE)	Free	Advance replacement with UI Care	~4%	Community-driven; no enterprise TAC.
MikroTik	Warranty	Via distributors/consultants	Free	Distributor RMA	~0–3%	No vendor TAC.

AMC (Annual Maintenance Contract) in India is often delivered by the system integrator on top of OEM support at 6–12% of hardware value per year, covering on-site engineers, spares and preventive maintenance; budget both OEM back-to-back support and SI AMC.

9. India Pricing & Market

9.1 Cost build-up for imported network equipment

Component	Typical value	Comment / source
Exchange rate	≈ ₹95 per USD (Sept 2026 mid-market)	Rupee weakened through 2026; build 3–5% FX buffer into multi-year INR budgets.
Basic Customs Duty (BCD)	0%–20% depending on HSN classification (e.g., 8517 62 / 8517 69 / 8517 18)	The Union Budget 2025–26 halved duty on carrier-grade Ethernet switches and rationalised BCD slabs; classification disputes (e.g., Ingram Micro/Cisco Catalyst 3850, CESTAT Jan 2025) remain common.
Social Welfare Surcharge / AIDC	Up to 10% of BCD	Applies on top of BCD where levied.
IGST / GST	18%	Recoverable as input tax credit for GST-registered enterprises; a real cost for exempt entities (some government, education, healthcare).
Distributor + SI margin	8–20% combined	Major distributors: Ingram Micro India, Redington, Savex, iValue, Rashi; SIs: TCS, Wipro, HCLTech, Tech Mahindra, Dimension Data/NTT, Sify, Orange Business, Comtel, Locuz, Embee.
Enterprise discount off global list	40–70% on large deals	Indian deal registrations frequently receive deeper discounts than US/EU; ask for India-specific BOQ pricing.
Lead time	2–6 weeks for stocked SKUs; 8–16 weeks for DC/high-end; Wi-Fi 7/800G may be longer	Make-in-India manufactured SKUs (Cisco, HPE/Aruba, Juniper and others have local manufacturing via EMS partners) shorten lead time and help PMA/GeM eligibility.
Government / PSU procurement	GeM portal, L1 tendering, PMA (Preferential Market Access) and Make-in-India local-content norms; Trusted Source rules for telecom	Huawei/ZTE are not designated trusted sources for telecom networks; many government bodies avoid Chinese vendors entirely.

9.2 India market analysis

Dimension	Assessment
Most commonly deployed (enterprise)	Cisco (clear #1 installed base), HPE Aruba, Fortinet (firewall/SD-WAN #1 by units), Juniper, Palo Alto, Check Point; Ruckus in hospitality/education; D-Link, TP-Link Omada, Netgear in SMB.
Telecom	Nokia, Ericsson, Samsung (RAN), Cisco, Juniper, Ciena (optical/IP); Huawei/ZTE limited to legacy since 2021 trusted-source mandate.
BFSI	Cisco + Palo Alto/Check Point/Fortinet dominate (RBI cybersecurity framework, SEBI CSCRF); Arista growing in DC; Aruba/Juniper Mist in branch WLAN.
Government	Cisco, HPE, Juniper; growing preference for Make-in-India and indigenous (e.g., C-DOT, Tejas, Sterlite/STL) in telecom/defence.
Manufacturing	Cisco IE (Industrial Ethernet), Fortinet OT security, HPE Aruba, Siemens/Moxa/Hirschmann for OT.
IT/ITES	Cisco, Arista (DC), Palo Alto, Zscaler (SSE), Fortinet; GCCs often mirror their HQ vendor standards.
Partner ecosystem	Deepest: Cisco > HPE Aruba ≈ Fortinet > Juniper > Palo Alto > Arista > Extreme.
Certified engineers	CCNA/CCNP very abundant; Fortinet NSE strong; Juniper JNCIA/JNCIS moderate; Aruba ACMP moderate; Arista ACE scarce.
Support quality	Cisco/Juniper/HPE have India TAC centres (Bengaluru); 4-hr depot coverage strong in Tier-1 cities, weaker in Tier-3.
Training	Abundant Cisco/Fortinet academies; Juniper and Aruba education partners in major metros.

9.3 Vendors with lower TCO in India

- **Fortinet** — single FortiGate licence covering NGFW + SD-WAN, FortiSwitch/FortiAP without per-device cloud fees when managed from FortiGate; strong local partner base.
- **HPE Aruba (Instant On for SMB; CX/AOS-10 for enterprise)** — lifetime hardware warranty on many models reduces AMC; aggressive India discounting.
- **Juniper Mist** — lower operational labour through AI-driven troubleshooting; competitive pricing in BFSI WLAN tenders.
- **Ruckus, D-Link/TP-Link Omada, Ubiquiti** — for SMB, hospitality and education where enterprise TAC is not mandatory.
- Labour is cheaper in India (modelled at ~25% of US cost), so subscriptions and hardware form a larger share of TCO than in the US — making licence negotiation the biggest lever.

9.4 Scenario B (500-user campus) — regional 5-year TCO

Vendor stack	US 5Y TCO (USD)	India 5Y TCO (INR, excl. GST)	India (USD equiv.)	Europe 5Y TCO (EUR, excl. VAT)	Europe (USD equiv.)
Cisco	\$763,713	₹487.8 lakh	\$513,427	€621,309	\$714,149
HPE (Aruba + Juniper)	\$645,501	₹404.2 lakh	\$425,497	€522,327	\$600,376
Arista (+ partner NGFW/SSE)	\$672,607	₹438.2 lakh	\$461,298	€549,285	\$631,363
Fortinet	\$523,206	₹280.4 lakh	\$295,107	€409,175	\$470,316
Extreme (+ FortiGate)	\$592,850	₹335.3 lakh	\$352,947	€468,902	\$538,967
Huawei*	\$529,379	₹260.8 lakh	\$274,553	€407,114	\$467,947
Ubiquiti	\$437,545	₹132.2 lakh	\$139,120	€312,067	\$358,697

Regional factors used: India — hardware ×1.12 (duty/logistics net of deeper discount), subscriptions ×0.95, labour ×0.25; Europe — hardware ×1.05, subscriptions ×1.05, labour ×0.80. GST/VAT excluded because they are normally recoverable. Market estimates, not quotations.

10. Europe Pricing & Market

Dimension	Assessment
EUR pricing	Most vendors publish EUR price lists to partners; EUR list is often ~equal to USD list numerically, i.e., ~10–15% dearer after FX. Example: Aruba Central Advanced 3-yr (JZ526AAE) €702 street (MSRP €890) vs \$766 (MSRP \$950) in the US (SHI, Oct 2026).
VAT	17% (LU) to 27% (HU); 19% DE, 20% FR/UK, 21% NL/ES, 22% IT. Recoverable for VAT-registered businesses.
Import/logistics	Most network equipment is 0% duty under the WTO ITA; main cost is intra-EU logistics and WEEE compliance.
GDPR / data residency	Cloud-managed platforms (Meraki, Central, Mist, CVaaS, ExtremeCloud IQ, FortiCloud) all offer EU-hosted regions; verify telemetry/PII processing locations and sub-processors. SASE PoPs within the EU and EU-only log storage are now a standard tender requirement.
Regulation	NIS2 (in force Oct 2024) raises security and supply-chain obligations for essential/important entities; EU 5G Toolbox — the Commission considers Member State decisions to restrict Huawei/ZTE justified; Cyber Resilience Act obligations phase in from 2026–2027.
Enterprise adoption	Cisco #1; HPE Aruba strong (DE, UK, NL, Nordics); Juniper strong in SP and finance; Extreme strong in UK public sector/education; Fortinet #1 in firewall units; Huawei material in some southern/eastern EU enterprises but declining in critical sectors.
Telecom	Nokia and Ericsson dominate RAN/IP; Cisco and Juniper in core/edge routing; Ciena and Nokia/Infinera in optical.
Data center	Arista and Cisco lead; Juniper Apstra and Nokia SR Linux/EDA gaining in sovereign cloud and SP DCs.
Local support / partners	Dense partner networks for Cisco, HPE, Fortinet, Palo Alto; country-specific depots generally provide 4-hr coverage in major cities.
Cost competitiveness	Fortinet and Extreme are frequently lowest in public tenders; Cisco wins on standardisation and skills; labour costs (~80% of US) make AIOps savings material.

10.1 Country-specific notes

- **Germany:** strict works-council and data-protection reviews for cloud management; preference for on-prem options (Catalyst Center, Aruba Central On-Premises, Apstra).
- **UK:** Huawei removal from 5G by 2027; public sector frameworks (Crown Commercial Service) favour multiple vendors.
- **France:** SecNumCloud qualification increasingly requested for SASE/cloud services in public and critical sectors.
- **Nordics/Netherlands:** high cloud-management adoption; strong Juniper/Arista presence in SP and DC.

11. US Pricing & Market

Dimension	Assessment
USD pricing	Reference market for global list prices; strongest public reseller pricing transparency (CDW, SHI, Connection, Insight).
Sales tax	State/local 0~10%; many enterprises are exempt or self-assess use tax; tariffs in 2025-2026 added surcharges on some imported hardware (e.g., Ubiquiti 'surcharge incl.' pricing: Pro Max 48 PoE \$1,299 → \$1,401).
Enterprise adoption	Cisco dominant; HPE Aruba + Juniper is the principal campus challenger; Extreme strong in education/SLED; Meraki very strong mid-market.
Data center	Arista leads cloud titan and high-end enterprise DC; Cisco Nexus/ACI strong in enterprise; NVIDIA Spectrum-X a new entrant in AI back-end fabrics.
Cloud	AWS/Azure/GCP native networking + Aviatrix/Alkira for multi-cloud; SASE from Zscaler, Palo Alto, Netskope, Cisco, Fortinet, Cato.
Government	FedRAMP, TAA, NDAA §889 (prohibits Huawei/ZTE), Common Criteria, FIPS 140-3: Cisco, Juniper, Arista, HPE, Palo Alto, Fortinet hold broad certifications.
Healthcare	Cisco, HPE Aruba, Juniper Mist, Extreme; HIPAA-driven segmentation and NAC (ISE/ClearPass).
Financial services	Cisco, Arista (low-latency trading), Juniper, Palo Alto, Zscaler.
Telecom	Cisco, Juniper (HPE), Nokia, Ciena, Arista; Huawei/ZTE excluded under FCC rip-and-replace.
Support & partners	Largest partner density and depot coverage for all US vendors.
Pricing competitiveness	Highest absolute labour cost makes operational efficiency (AIOps, automation) the dominant TCO lever.

12. 3-Year TCO

TCO = Hardware + Software + Licensing + Support + Maintenance + Management + Security + Optics + Installation + Professional Services + Training + Migration + Operational Costs.

12.1 Model assumptions

Assumption	Value used
Price basis	USD estimated street/partner prices (Market estimate) calibrated against the public reseller prices in Sections 6–7; US pricing, excluding tax.
Capex components	Hardware + optics (2% branch; 7–12% campus/DC by vendor) + spares (5%) + installation/professional services/migration (10–20% of hardware) + training (one-off) + management platform (Scenarios C/D).
Opex components	Licences, cloud management, support/AMC, security subscriptions, NAC per endpoint, SASE per user, and operations labour.
Escalation	4% per year on subscriptions/support from Year 2.
Operations labour	Network-engineer FTE at \$140,000/yr fully loaded (US), multiplied by an operational-efficiency factor: Cisco 1.00, HPE 0.88 (Mist/Central AI Ops), Arista 0.85, Fortinet 0.90, Extreme 0.95, Huawei 1.00, Ubiquiti 1.15.
Stacks	Arista and Extreme stacks include partner NGFW/SSE (FortiGate or Palo Alto) because they lack first-party NGFW. Huawei* shown for reference only. Ubiquiti lacks enterprise NAC, SASE and DC — modelled only for A and B and is not like-for-like.
Not included	WAN/DIA circuits (same for every vendor), power/cooling, hardware refresh beyond 5 years, Cisco-to-X retraining beyond the training line.

Scenario A – Small branch (50 users)

Bill of materials: 1× branch SD-WAN/NGFW, 2× 24-port PoE access switch, 3× Wi-Fi 7 AP; cloud management. Operations labour: 0.05 FTE × \$140,000 adjusted by vendor operational-efficiency factor.

Vendor stack	Initial CAPEX	Year 1 OPEX	Year 2 OPEX	Year 3 OPEX	Year 4 OPEX	Year 5 OPEX	3-Year TCO	5-Year TCO
Cisco	\$11,614	\$8,700	\$8,768	\$8,839	\$8,912	\$8,989	\$37,921	\$55,822
HPE (Aruba + Juniper)	\$11,250	\$7,750	\$7,814	\$7,880	\$7,949	\$8,020	\$34,693	\$50,662
Arista (+ partner NGFW/SSE)	\$12,668	\$7,900	\$7,978	\$8,059	\$8,143	\$8,231	\$36,605	\$52,980
Fortinet	\$7,875	\$7,290	\$7,330	\$7,371	\$7,414	\$7,458	\$29,865	\$44,737
Extreme (+ FortiGate)	\$10,509	\$7,970	\$8,023	\$8,078	\$8,135	\$8,194	\$34,580	\$50,909
Huawei*	\$7,875	\$7,630	\$7,655	\$7,681	\$7,709	\$7,737	\$30,842	\$46,287
Ubiquiti	\$4,127	\$8,110	\$8,112	\$8,115	\$8,117	\$8,120	\$28,464	\$44,702

Vendor stack	Hardware (excl. services)	Annual subs + support (Y1)	Annual ops labour	5Y TCO per site	5Y TCO per user	5Y TCO per device	Annualised TCO
Cisco	\$8,700	\$1,700	\$7,000	\$55,822	\$1,116	\$9,304	\$11,164
HPE (Aruba + Juniper)	\$7,400	\$1,590	\$6,160	\$50,662	\$1,013	\$8,444	\$10,132
Arista (+ partner NGFW/SSE)	\$8,400	\$1,950	\$5,950	\$52,980	\$1,060	\$8,830	\$10,596
Fortinet	\$5,100	\$990	\$6,300	\$44,737	\$895	\$7,456	\$8,947

Vendor stack	Hardware (excl. services)	Annual subs + support (Y1)	Annual ops labour	5Y TCO per site	5Y TCO per user	5Y TCO per device	Annualised TCO
Extreme (+ FortiGate)	\$6,700	\$1,320	\$6,650	\$50,909	\$1,018	\$8,485	\$10,182
Huawei*	\$4,700	\$630	\$7,000	\$46,287	\$926	\$7,715	\$9,257
Ubiquiti	\$3,100	\$60	\$8,050	\$44,702	\$894	\$7,450	\$8,940

Lowest 5-year TCO in this scenario: **Ubiquiti** (\$45k); highest: **Cisco** (\$56k). Cisco vs. lowest: +25%.

Scenario B – Medium campus (500 users)

Bill of materials: 2× campus NGFW/SD-WAN, 10× 48-port PoE access switch, 2× core/distribution switch, 25× Wi-Fi 7 AP; NAC for 1,000 endpoints; cloud management. Operations labour: 0.5 FTE × \$140,000 adjusted by vendor operational-efficiency factor.

Vendor stack	Initial CAPEX	Year 1 OPEX	Year 2 OPEX	Year 3 OPEX	Year 4 OPEX	Year 5 OPEX	3-Year TCO	5-Year TCO
Cisco	\$197,060	\$110,000	\$111,600	\$113,264	\$114,995	\$116,794	\$531,924	\$763,713
HPE (Aruba + Juniper)	\$169,595	\$92,600	\$93,840	\$95,130	\$96,471	\$97,866	\$451,165	\$645,501
Arista (+ partner NGFW/SSE)	\$186,890	\$94,250	\$95,640	\$97,086	\$98,589	\$100,153	\$473,866	\$672,607
Fortinet	\$113,420	\$80,500	\$81,200	\$81,928	\$82,685	\$83,473	\$357,048	\$523,206
Extreme (+ FortiGate)	\$137,941	\$89,100	\$90,004	\$90,944	\$91,922	\$92,939	\$407,989	\$592,850
Huawei*	\$103,550	\$84,000	\$84,560	\$85,142	\$85,748	\$86,378	\$357,252	\$529,379
Ubiquiti	\$32,012	\$81,060	\$81,082	\$81,106	\$81,130	\$81,155	\$275,260	\$437,545

Vendor stack	Hardware (excl. services)	Annual subs + support (Y1)	Annual ops labour	5Y TCO per site	5Y TCO per user	5Y TCO per device	Annualised TCO
Cisco	\$145,500	\$40,000	\$70,000	\$763,713	\$1,527	\$19,582	\$152,743
HPE (Aruba + Juniper)	\$121,500	\$31,000	\$61,600	\$645,501	\$1,291	\$16,551	\$129,100
Arista (+ partner NGFW/SSE)	\$133,000	\$34,750	\$59,500	\$672,607	\$1,345	\$17,246	\$134,521
Fortinet	\$82,000	\$17,500	\$63,000	\$523,206	\$1,046	\$13,416	\$104,641
Extreme (+ FortiGate)	\$97,700	\$22,600	\$66,500	\$592,850	\$1,186	\$15,201	\$118,570
Huawei*	\$73,500	\$14,000	\$70,000	\$529,379	\$1,059	\$13,574	\$105,876
Ubiquiti	\$25,010	\$560	\$80,500	\$437,545	\$875	\$11,219	\$87,509

Lowest 5-year TCO in this scenario: **Ubiquiti** (\$438k); highest: **Cisco** (\$764k). Cisco vs. lowest: +75%.

13. 5-Year TCO

The 5-year tables above (Scenarios A and B) and below (Scenarios C and D) show CAPEX + Year 1–5 OPEX. Scenarios C and D are where licensing and SASE/NAC subscriptions dominate.

Scenario C – Large enterprise (5,000 users, 3 campuses + DC + 10 branches)

Bill of materials: 6× campus NGFW/SD-WAN, 10× branch SD-WAN/NGFW, 110× 48-port PoE access switch, 6× core/distribution switch, 8× DC leaf, 2× DC spine (400G), 2× DC firewall, 220× Wi-Fi 7 AP; NAC for 10,000 endpoints; SASE/SSE for 5,000 users; on-prem/hybrid management platform. Operations labour: 4 FTE × \$140,000 adjusted by vendor operational-efficiency factor.

Vendor stack	Initial CAPEX	Year 1 OPEX	Year 2 OPEX	Year 3 OPEX	Year 4 OPEX	Year 5 OPEX	3-Year TCO	5-Year TCO
Cisco	\$1,989,880	\$1,512,000	\$1,550,080	\$1,589,683	\$1,630,871	\$1,673,705	\$6,641,643	\$9,946,219
HPE (Aruba + Juniper)	\$1,643,070	\$1,281,400	\$1,312,944	\$1,345,750	\$1,379,868	\$1,415,350	\$5,583,164	\$8,378,382
Arista (+ partner NGFW/SSE)	\$1,764,100	\$1,416,000	\$1,453,600	\$1,492,704	\$1,533,372	\$1,575,667	\$6,126,404	\$9,235,443
Fortinet	\$1,195,050	\$1,125,600	\$1,150,464	\$1,176,323	\$1,203,215	\$1,231,184	\$4,647,437	\$7,081,836
Extreme (+ FortiGate)	\$1,370,528	\$1,274,700	\$1,304,408	\$1,335,304	\$1,367,436	\$1,400,854	\$5,284,940	\$8,053,231
Huawei*	\$1,015,800	\$1,201,300	\$1,226,952	\$1,253,630	\$1,281,375	\$1,310,230	\$4,697,682	\$7,289,288

Vendor stack	Hardware (excl. services)	Annual subs + support (Y1)	Annual ops labour	5Y TCO per site	5Y TCO per user	5Y TCO per device	Annualised TCO
Cisco	\$1,409,000	\$952,000	\$560,000	\$710,444	\$1,989	\$27,325	\$1,989,244
HPE (Aruba + Juniper)	\$1,179,000	\$788,600	\$492,800	\$598,456	\$1,676	\$23,018	\$1,675,676
Arista (+ partner NGFW/SSE)	\$1,270,000	\$940,000	\$476,000	\$659,675	\$1,847	\$25,372	\$1,847,089
Fortinet	\$855,000	\$621,600	\$504,000	\$505,845	\$1,416	\$19,456	\$1,416,367
Extreme (+ FortiGate)	\$981,600	\$742,700	\$532,000	\$575,231	\$1,611	\$22,124	\$1,610,646
Huawei*	\$716,000	\$641,300	\$560,000	\$520,663	\$1,458	\$20,026	\$1,457,858

Lowest 5-year TCO in this scenario: **Fortinet** (\$7.08M); highest: **Cisco** (\$9.95M). Cisco vs. lowest: +40%.

Scenario D – Global enterprise (10,000 users, 50 sites, India/EU/US, 2 DCs)

Bill of materials: 10× campus NGFW/SD-WAN, 45× branch SD-WAN/NGFW, 300× 48-port PoE access switch, 100× 24-port PoE access switch, 16× core/distribution switch, 24× DC leaf, 4× DC spine (400G), 4× DC firewall, 600× Wi-Fi 7 AP; NAC for 20,000 endpoints; SASE/SSE for 10,000 users; on-prem/hybrid management platform. Operations labour: 10 FTE × \$140,000 adjusted by vendor operational-efficiency factor.

Vendor stack	Initial CAPEX	Year 1 OPEX	Year 2 OPEX	Year 3 OPEX	Year 4 OPEX	Year 5 OPEX	3-Year TCO	5-Year TCO
Cisco	\$5,259,040	\$3,472,000	\$3,554,880	\$3,641,075	\$3,730,718	\$3,823,947	\$15,926,995	\$23,481,660
HPE (Aruba + Juniper)	\$4,360,165	\$2,944,250	\$3,012,740	\$3,083,970	\$3,158,048	\$3,235,090	\$13,401,125	\$19,794,263
Arista (+ partner NGFW/SSE)	\$4,709,290	\$3,198,500	\$3,278,840	\$3,362,394	\$3,449,289	\$3,539,661	\$14,549,024	\$21,537,974

Vendor stack	Initial CAPEX	Year 1 OPEX	Year 2 OPEX	Year 3 OPEX	Year 4 OPEX	Year 5 OPEX	3-Year TCO	5-Year TCO
Fortinet	\$3,170,555	\$2,570,750	\$2,623,180	\$2,677,707	\$2,734,415	\$2,793,392	\$11,042,192	\$16,570,000
Extreme (+ FortiGate)	\$3,699,770	\$2,914,000	\$2,977,360	\$3,043,254	\$3,111,785	\$3,183,056	\$12,634,384	\$18,929,225
Huawei*	\$2,665,200	\$2,735,050	\$2,788,452	\$2,843,990	\$2,901,750	\$2,961,820	\$11,032,692	\$16,896,261

Vendor stack	Hardware (excl. services)	Annual subs + support (Y1)	Annual ops labour	5Y TCO per site	5Y TCO per user	5Y TCO per device	Annualised TCO
Cisco	\$3,772,000	\$2,072,000	\$1,400,000	\$469,633	\$2,348	\$21,289	\$4,696,332
HPE (Aruba + Juniper)	\$3,150,500	\$1,712,250	\$1,232,000	\$395,885	\$1,979	\$17,946	\$3,958,853
Arista (+ partner NGFW/SSE)	\$3,413,000	\$2,008,500	\$1,190,000	\$430,759	\$2,154	\$19,527	\$4,307,595
Fortinet	\$2,290,500	\$1,310,750	\$1,260,000	\$331,400	\$1,657	\$15,023	\$3,314,000
Extreme (+ FortiGate)	\$2,669,000	\$1,584,000	\$1,330,000	\$378,584	\$1,893	\$17,162	\$3,785,845
Huawei*	\$1,904,000	\$1,335,050	\$1,400,000	\$337,925	\$1,690	\$15,318	\$3,379,252

Lowest 5-year TCO in this scenario: **Fortinet** (\$16.57M); highest: **Cisco** (\$23.48M). Cisco vs. lowest: +42%.

13.1 Interpretation

- **Lowest hardware price ≠ lowest TCO.** Ubiquiti has the lowest capex in A/B by a wide margin, but its 5-year TCO advantage narrows because operational labour dominates and it lacks NAC, SASE and enterprise TAC.
- **Subscriptions dominate from the medium campus upward.** In Scenario C, recurring licences, support, NAC and SASE are ~45–55% of the 5-year TCO for every vendor (rising to ~80–85% once operations labour is added); negotiating renewal caps is worth more than the upfront discount.
- **Cisco premium** is roughly 25–45% versus the lowest enterprise-grade stack (Fortinet) and ~15–18% versus HPE; it buys skills availability, ecosystem breadth and lowest migration risk.
- **Fortinet** is consistently lowest among enterprise-grade options because security, SD-WAN and LAN/WLAN management share one licence and OS — but its data-center fabric is weaker, so Scenario D assumes a less capable DC design.
- **HPE Networking** sits roughly mid-way between Fortinet and Cisco while offering the closest functional parity with Cisco.

13.2 Regional 5-year TCO — Scenario C (5,000 users)

Vendor stack	US 5Y TCO (USD)	India 5Y TCO (INR, excl. GST)	India (USD equiv.)	Europe 5Y TCO (EUR, excl. VAT)	Europe (USD equiv.)
Cisco	\$9,946,219	₹7432.4 lakh	\$7,823,588	€8,475,566	\$9,742,030
HPE (Aruba + Juniper)	\$8,378,382	₹6183.2 lakh	\$6,508,585	€7,115,774	\$8,179,051
Arista (+ partner NGFW/SSE)	\$9,235,443	₹7031.5 lakh	\$7,401,568	€7,916,752	\$9,099,715
Fortinet	\$7,081,836	₹4904.6 lakh	\$5,162,703	€5,919,635	\$6,804,178
Extreme (+ FortiGate)	\$8,053,231	₹5715.4 lakh	\$6,016,159	€6,776,119	\$7,788,642
Huawei*	\$7,289,288	₹4875.5 lakh	\$5,132,109	€6,047,807	\$6,951,502

14. Vendor Scorecard

Category	Weight
Technical capability	20%
Performance	10%
Security	10%
SD-WAN/SASE	10%
Cloud management	10%
Automation/API	5%
Reliability	10%
Licensing	5%
5-Year TCO	10%
Support ecosystem	5%
Global availability	5%

Rank	Vendor	Tech	Perf	Sec	SD-WAN/SASE	Cloud	Auto	Rel	Lic	TCO	Supp	Global	Weighted
1	Cisco	9	9	8	8	8	8	9	4	5	10	10	8.10
2	Fortinet	7	8	9	9	8	7	8	8	9	7	9	8.05
3	HPE Aruba Networking	8	8	7	8	9	8	8	6	7	8	9	7.85
4	Juniper (HPE)	8	9	7	7	9	9	9	6	6	7	8	7.80
5	Arista Networks	8	10	5	6	7	10	10	7	6	7	8	7.60
6	Huawei	8	9	7	7	8	7	8	7	9	6	4	7.60
7	Palo Alto Networks	6	8	10	8	8	8	9	4	4	8	9	7.35
8	Cato Networks (added)	4	7	9	9	9	7	8	7	7	7	8	7.15
9	Versa Networks	6	7	8	9	7	7	7	7	7	6	7	7.05
10	Extreme Networks	7	7	6	5	8	7	8	8	7	7	7	6.95
11	Check Point	5	8	9	5	6	6	8	5	5	7	8	6.40
12	Zscaler (added)	3	7	9	4	9	7	8	6	6	8	9	6.40
13	Nokia	6	9	5	4	5	8	9	6	6	6	6	6.30
14	Ruckus Networks	6	6	4	2	7	5	7	7	8	6	7	5.85
15	Ciena	5	9	4	2	4	8	9	6	6	6	7	5.75
16	VMware by Broadcom	6	7	6	5	5	7	8	2	3	5	7	5.65
17	Dell Technologies	5	7	3	2	4	6	7	7	8	6	7	5.40
18	Ubiquiti	4	5	3	2	7	3	6	10	10	3	6	5.20
19	MikroTik	4	5	3	2	3	4	5	10	10	2	5	4.65

Scores are relative to full-stack enterprise needs. Security/SASE specialists (Palo Alto, Check Point, Cato, Zscaler) and SP/optical vendors (Nokia, Ciena) score lower on 'technical capability' because they do not cover campus/DC — they may still be the best choice in their category.

14.1 Why each vendor received its score

Vendor	Weighted	Rationale
Cisco	8.10	Broadest, deepest portfolio; best support/skills/global reach; penalised on licensing complexity and TCO.
Fortinet	8.05	Security + SD-WAN + LAN on one OS and licence; top TCO and SD-WAN scores; weaker DC fabric and some CVE history.
HPE Aruba Networking	7.85	Leader in campus/WLAN, NAC and cloud mgmt (Central); EdgeConnect SD-WAN; integration with Juniper still maturing.
Juniper (HPE)	7.80	Best-in-class AIOps (Mist/Marvis), strong routing and DC (Apstra); post-merger roadmap rationalisation risk.
Arista Networks	7.60	Top DC performance, reliability and automation; campus and VeloCloud SD-WAN still building scale; no first-party NGFW.
Huawei	7.60	Technically strong and low cost; global availability score depressed by regulatory restrictions in US/EU/India telecom.
Palo Alto Networks	7.35	Best-in-class security and SASE; not a LAN/DC vendor; premium subscription pricing.
Cato Networks (added)	7.15	Clean single-vendor SASE with private backbone; not a LAN/DC vendor.
Versa Networks	7.05	Strong unified SD-WAN/SASE, popular with service providers; smaller enterprise footprint.
Extreme Networks	6.95	Simple pooled licensing, good fabric and cloud management, good TAC; smaller ecosystem; partner NGFW needed.
Check Point	6.40	Strong threat prevention and management; limited networking breadth.
Zscaler (added)	6.40	SSE leader; not a network-hardware vendor (low tech-breadth score by design).
Nokia	6.30	Excellent SP routing/DC (SR Linux, EDA); limited enterprise campus/WLAN.
Ruckus Networks	5.85	Good RF performance and value; limited portfolio; ownership changes under CommScope restructuring.
Ciena	5.75	Optical/SP leader; not an enterprise LAN/WLAN vendor.
VMware by Broadcom	5.65	NSX remains capable but licensing changes (VCF-only, per-core) and SD-WAN divestiture lower the score.
Dell Technologies	5.40	Cost-effective open networking (SONiC) for DC; little campus/security.
Ubiquiti	5.20	Excellent price and simplicity for SMB; lacks enterprise TAC, NAC, SASE, DC.
MikroTik	4.65	Extremely low cost and flexible RouterOS; steep learning curve, no enterprise TAC, security track record concerns.

15. Cisco Alternatives

For each competitor: **if an organisation currently uses Cisco, what does it gain and lose by migrating?** Migration difficulty: 1 = trivial, 10 = very hard.

Vendor	What you gain	What you lose	CLI / skills	APIs / automation	Interop with remaining Cisco	Licensing / TCO impact	Migration difficulty
HPE Aruba (+ Juniper)	Strong WLAN/NAC, Central AIops, lifetime warranty, lower TCO (~15%)	SD-Access/SGT ecosystem, Cisco breadth, some CCNP familiarity	AOS-CX CLI is Cisco-like; retrain ~2-4 weeks	REST, Ansible, Terraform; good	Good (standards-based; ClearPass works with Cisco)	Simpler; Central per device	5
Juniper (HPE)	Mist AI, Junos stability, Apstra, SP routing	Familiar IOS syntax; EIGRP; ACI	Junos commit model differs; 4-8 weeks	Excellent (NETCONF native, PyEZ)	Good; BGP/OSPF interop	Mist subs per device; ~10-20% lower TCO	6
Arista	DC performance, CloudVision, software quality	Integrated NGFW, Cisco campus breadth	EOS CLI very close to IOS/NX-OS; 1-2 weeks	Excellent (eAPI, gNMI, AVD)	Excellent in DC (EVPN, MLAG)	Simpler; often lower DC TCO	4 (DC) / 6 (campus)
Fortinet	Lowest TCO, integrated security + SD-WAN	DC fabric, advanced campus features, ISE depth	FortiOS GUI-centric; NSE certification	REST, Ansible, Terraform	Good at WAN edge; FortiSwitch only with FortiGate	Lowest; fewer SKUs	5
Extreme	Pooled licensing, fabric simplicity, good TAC	Ecosystem, skills availability, integrated security	EXOS/VOSS CLI differ; 3-6 weeks	REST/Ansible; fewer Terraform resources	Good (standards)	Simpler; ~15-25% lower TCO	6
Huawei	Price, broad portfolio	Regulatory acceptability, Western partner depth	VRP CLI resembles Cisco; 2-4 weeks	NETCONF/RESTCONF	Good (standards)	Lowest hardware cost	5 technical / 9 compliance
Palo Alto (security/SASE)	Best threat prevention, Prisma SASE	Cisco Secure Firewall/ISE integration	PAN-OS GUI/CLI; PCNSE	Strong APIs, Terraform	Good; pairs with any LAN	Higher security spend	4
Check Point (security)	Strong prevention, mature mgmt	Networking breadth	Gaia CLI/SmartConsole	Good APIs	Good	Mid-to-high	5
Versa (SD-WAN/SASE)	Unified SASE, multi-tenancy	Cisco SD-WAN/Meraki familiarity	New UI (Director)	REST	Good at WAN edge	Often lower via SP	6
Nokia (SP/DC)	Routing scale, model-driven DC	Enterprise breadth	SR OS/SR Linux very different; 8+ weeks	Excellent (gNMI-first)	Good (standards)	Neutral	8
Dell (DC)	Open networking cost	Ecosystem, TAC depth	SONiC/OS10 Linux-like	Good (SONiC)	Good (EVPN)	Lower	6
Ubiquiti / MikroTik (SMB)	Very low cost	Enterprise support, NAC, HA	GUI (UniFi) / RouterOS	Limited	Basic	Lowest capex	3 (SMB) / 8 (enterprise)

16. Migration Analysis

16.1 Migration complexity drivers

Driver	Low complexity	High complexity
Proprietary protocols	OSPF/BGP/IS-IS everywhere	EIGRP, PVST+, VTP, SGT/TrustSec, ACI contracts
Security integration	Standalone NAC/firewall	ISE-driven dynamic segmentation, Stealthwatch, Umbrella tie-ins
Operations tooling	Generic NMS/SNMP, Ansible	Catalyst Center templates, Meraki-specific APIs, SolarWinds custom scripts
Skills	Teams with multi-vendor certs	CCNP-only teams, no automation
Contracts	Support co-terminating soon	EA/ELA with true-forward; long DNA subscriptions

16.2 Recommended migration approach

1. **Discover:** inventory devices, licences, contract end-dates, proprietary features (EIGRP, TrustSec, VTP).
2. **Normalise:** migrate EIGRP → OSPF/BGP, PVST+ → MSTP, and SGT → standards-based (VXLAN-GBP / RADIUS VLAN) before swapping hardware.
3. **Pilot:** one branch + one campus building; validate NAC, wireless roaming, QoS, multicast.
4. **Phase by refresh cycle:** replace at EoL or contract expiry to avoid paying twice; keep Cisco core during access-layer migration (interop is standards-based).
5. **Retrain and automate:** budget 5–10% of capex for training and automation (Ansible/Terraform) to realise the opex savings modelled.
6. **Exit cleanly:** confirm Smart Account licence termination, return-to-vendor credits, and data-wipe certificates.

Typical migration professional-services cost: 10–20% of new hardware value for campus, 15–25% for DC/SD-WAN; duration 3–6 months (campus of 500 users) to 18–36 months (global enterprise).

17. Vendor Strengths and Weaknesses

Cisco

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Broadest portfolio across every category	Highest TCO and complex licensing	Global enterprise	Cost-sensitive SMB
Largest skills pool (CCNA/CCNP/CCIE)	Mandatory term subscriptions on Catalyst	Regulated BFSI/government	Greenfield cloud-native DC wanting simplicity
Global TAC and depot coverage	Two campus stacks (Catalyst vs Meraki) still converging	Large campus with existing Cisco skills	Organisations wanting simple licensing
Security + observability ecosystem (Splunk, ThousandEyes)	Frequent product renaming/EoL churn		
Mature SD-Access, ACI, Catalyst SD-WAN	Vendor lock-in via proprietary features (EIGRP, SGT, ACI)		

Key differentiator: The safest single-vendor end-to-end choice with the deepest ecosystem. **Main competitor:** HPE Networking.

HPE Aruba Networking

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Excellent WLAN and Central cloud management	Two cloud platforms (Central and Mist) to rationalise	Campus/WLAN	Tier-1 SP core
ClearPass NAC (multi-vendor)	Smaller DC footprint (pre-Juniper)	Healthcare and education	Pure security-led branch
EdgeConnect SD-WAN with WAN optimisation	SSE less mature than leaders	Mid-to-large enterprise	Very small SMB without partner
Lifetime warranty on many switches	Gateway licensing complexity		
Now backed by Juniper routing/DC/security	Partner depth thinner than Cisco in some regions		

Key differentiator: AI-driven campus and NAC with Juniper's routing and DC under one owner. **Main competitor:** Cisco.

Juniper (HPE)

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Mist AI/Marvis assurance leadership	Post-merger roadmap uncertainty	AIOps-driven campus	Security-first branch
Junos reliability and automation (NETCONF native)	Smaller campus market share than Aruba/Cisco	Service provider	Ultra-low-budget SMB
Apstra multi-vendor intent-based DC	SRX less competitive than PAN/Fortinet	DC with multi-vendor intent automation	Shops dependent on Cisco-only features
SP-grade routing (MX/PTX)	Fewer certified engineers in India		
Session Smart SD-WAN (tunnel-free)	Pricing variability post-integration		

Key differentiator: The most mature AI-native operations model in networking. **Main competitor:** Cisco.

Arista Networks

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Highest DC performance and 400G/800G AI fabrics	No first-party NGFW	Data center / AI fabrics	Small branch with integrated security
Single EOS image, very low defect rate	Campus/WLAN share still small	Financial trading	SMB
CloudVision state streaming automation	VeloCloud integration in progress	Cloud and SP peering	Vertical needing large WLAN install base references
Cisco-like CLI eases migration	Fewer partners in India		
Strong financials and hyperscaler validation	Premium on optics/support in some regions		

Key differentiator: Software quality and DC automation that others benchmark against. **Main competitor:** Cisco.

Huawei

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Very competitive pricing	Regulatory exclusion (US NDAA §889, EU 5G Toolbox, India trusted source)	Price-sensitive enterprises in permitted markets	US federal/regulated
Full-stack portfolio incl. Wi-Fi 7 and 400G	Geopolitical and supply-chain risk	Campus/DC in Asia/MEA	EU critical infrastructure
Gartner WLAN Leader (2025)	Thin Western partner ecosystem	Large-scale WLAN	Indian telecom/government
Strong in Asia, MEA, LATAM	Limited SASE outside China		
iMaster NCE automation	Reputational risk in regulated sectors		

Key differentiator: Best price-for-capability where it is legally and politically acceptable. **Main competitor:** Cisco.

Nokia

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
SR OS / FP5 routing scale	Little enterprise campus/WLAN	Telecom/SP	Enterprise campus
SR Linux + EDA model-driven DC	Small enterprise channel	Utilities and rail	Branch SD-WAN
IP/optical convergence (Infinera)	Fewer enterprise engineers	Large DC fabrics with NetDevOps teams	SMB
Strong SP references	Limited SASE/security		
Private wireless (4G/5G) for industry	Higher complexity for enterprises		

Key differentiator: Carrier-grade routing and open model-driven DC automation. **Main competitor:** Cisco / Juniper.

Extreme Networks

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Pooled, portable Platform ONE licensing	Smaller ecosystem and skills pool	Education	SP core

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Fabric Connect (SPB) simplicity	No first-party NGFW/SSE	Healthcare	AI DC
Respected in-house TAC	Limited DC footprint	Mid-market campus	Security-led branch
Strong in education/SLED/healthcare	Lower brand recognition in India		
Cloud management with third-party device support	Fewer Terraform/automation resources		

Key differentiator: Simplest licensing model among the enterprise campus vendors. **Main competitor:** HPE Aruba.

Fortinet

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Single FortiOS across NGFW, SD-WAN, LAN, WLAN, SASE	Recurring critical CVEs on edge devices	Distributed branches/retail	Large DC fabric
ASIC-based price/performance	Weak DC fabric	Mid-market	Complex campus needing advanced WLAN analytics
Lowest TCO for secure branch	WLAN/switching less feature-rich than leaders	Cost-sensitive secure SD-WAN	SP routing
Largest firewall unit base	Support quality inconsistent at scale		
Strong channel in India/EU	Lock-in to Security Fabric		

Key differentiator: Security and networking converged on one OS and licence. **Main competitor:** Palo Alto Networks.

Palo Alto Networks

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Leading NGFW and threat prevention	Premium pricing; subscriptions 3–4x appliance over 5 yrs	BFSI security	Campus LAN
Prisma SASE (3x Gartner Leader, 2025)	No LAN/WLAN/DC switching	Global SASE	Budget SMB
Strong ZTNA 2.0, DLP, CASB	Licensing complexity	Zero trust programmes	DC fabric
Cortex XDR/XSIAM integration	Renewal escalation risk		
Platformisation discounts	Hardware refresh costs		

Key differentiator: Security depth from data center to SASE. **Main competitor:** Fortinet.

Check Point

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Strong threat prevention	Limited networking breadth	BFSI/government firewalls	Campus LAN
Centralised management (SmartConsole)	SD-WAN less mature	Multi-domain security mgmt	WLAN
Long BFSI/government track record	Management licences add cost	Existing Check Point estates	DC fabric

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Harmony SASE	Smaller SASE footprint		
Consistent policy model	Perceived complexity		

Key differentiator: Mature prevention-first security management. **Main competitor:** Palo Alto Networks.

Versa Networks

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Unified SD-WAN + SASE on one OS	Smaller company, private	Managed SD-WAN via SP	Campus LAN
Multi-tenancy for service providers	Complex Director UI	SASE for mid-large enterprise	DC fabric
Gartner SD-WAN Leader 5 years (to 2024)	Limited LAN/WLAN	Multi-tenant MSPs	Small SMB self-managed
Flexible deployment (on-prem/cloud)	Smaller partner base in India		
Competitive pricing	Fewer engineers		

Key differentiator: SP-friendly unified SASE stack. **Main competitor:** Fortinet.

VMware by Broadcom

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
NSX micro-segmentation in VCF	VCF-only, per-core subscription	Existing VCF private clouds	Physical campus/WLAN
Deep vSphere integration	Major price increases	Micro-segmentation in VMware estates	Cost-sensitive buyers
Avi load balancing	SD-WAN divested	Multi-tenant private cloud	Non-VMware workloads
Mature DFW	Partner programme cut-backs		
Large installed base	Customer trust erosion		

Key differentiator: Network virtualisation inside the VMware private cloud. **Main competitor:** Cisco ACI / Arista / cloud-native.

Ciena

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Coherent optics (WaveLogic) leadership	Not an enterprise LAN vendor	Optical transport	Campus
Metro/access routing	Limited security	DCI	WLAN
Navigator automation	Small enterprise channel	SP metro	SMB
Strong SP references	Specialised skills		
Reliability	Not relevant for campus		

Key differentiator: Optical networking specialist. **Main competitor:** Nokia.

Dell Technologies

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Enterprise SONiC open networking	Limited campus/WLAN/security	Open DC fabric	Campus
Cost-effective DC switching	Smaller networking TAC	HCI/AI clusters with Dell compute	Branch security
Integration with Dell servers/storage	SONiC skills scarce	Cost-sensitive DC	SP
800G platforms	Fewer automation integrations		
Bundled procurement	Lower brand recognition in networking		

Key differentiator: Open-networking DC for Dell infrastructure shops. **Main competitor:** Arista.

Ubiquiti

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Lowest acquisition cost	No enterprise TAC	SMB	Regulated enterprise
No licence fees	No NAC/SASE/DC	Small offices/retail	Large campus
Simple UI	Limited routing/HA	Hospitality (small)	DC
Wi-Fi 7 at low cost	Tariff surcharges		
Fast adoption	Scale limits		

Key differentiator: Enterprise-looking networking without subscriptions. **Main competitor:** TP-Link Omada / MikroTik.

Ruckus Networks

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Strong RF (BeamFlex)	Narrow portfolio	Hospitality	Global enterprise
Hospitality/MDU/education focus	No SD-WAN/security	Education	DC
RUCKUS One cloud	Ownership changes	Stadiums/venues	Security-led branch
Good value switching (ICX)	Smaller enterprise TAC		
IoT integration	Limited DC		

Key differentiator: RF performance in dense, challenging environments. **Main competitor:** HPE Aruba.

MikroTik

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Extremely low cost	No enterprise TAC	Small ISPs	Regulated enterprise

Top 5 strengths	Top 5 weaknesses	Best suited for (top 3)	Poor fit for (top 3)
Flexible RouterOS	Steep learning curve	Labs/test beds	Large campus
Good for ISPs/labs	Past security issues (botnets on unpatched devices)	Budget branch routing	Global enterprise
No licences	Limited HA/management		
Capable routing (BGP/OSPF/MPLS)	Compliance concerns		

Key differentiator: Maximum routing capability per dollar. **Main competitor:** Ubiquiti.

18. Industry Recommendations

Use case	Recommended vendor	Product family	Reason	Main advantage	Main disadvantage	Cost position
Small business	Fortinet / Ubiquiti	FortiGate 40F-70G + FortiAP; UniFi	Low capex, simple	One box security + SD-WAN	Limited TAC (Ubiquiti)	Low
Mid-market	HPE Aruba or Meraki (Cisco)	Central/Instant On; Meraki MS/MR/MX	Cloud mgmt, small IT teams	Fast deployment	Mandatory subscriptions	Medium
Large enterprise	Cisco or HPE Networking	Catalyst/ISE/SD-WAN; Aruba CX + Mist + ClearPass	Breadth, scale	Full stack	Cost / integration	High / Medium-high
Global enterprise	Cisco or HPE	As above + SSE	Global support	Consistent operations	Complex licensing	High
Data center	Arista	7050X/7060X/7800R + CloudVision	Performance, automation	Reliability	No NGFW	Medium-high
Campus LAN	HPE Aruba / Juniper Mist	CX 6000/6300, EX4100 + Mist	AI/ops, NAC	Lower ops effort	Roadmap convergence	Medium
Branch networking	Fortinet	FortiGate + FortiSwitch + FortiAP	Integrated security	Lowest TCO	Branch lock-in	Low
SD-WAN	Fortinet / Cisco / HPE EdgeConnect	FortiGate; Catalyst SD-WAN; EdgeConnect	Gartner Leaders	Mature, scalable	Licence tiers	Low-Medium
SASE	Palo Alto Prisma SASE / Fortinet FortiSASE / Cato	Prisma Access + SD-WAN	Security depth	Single-vendor SASE	Premium price (PAN)	Medium-high
Cloud networking	Cloud-native + Arista/Cisco/Aviatrix	AWS/Azure native, CloudVision, Catalyst SD-WAN Cloud OnRamp	Multi-cloud visibility	Consistent policy	Extra licences	Medium
Service provider	Nokia / Juniper / Cisco	7750 SR; MX/PTX; ASR 9000/8000	Routing scale	SR/SRv6, automation	High cost	High
Telecom (optical)	Ciena / Nokia (Infinera)	WaveLogic, 6500/Waveserver	Optical leadership	Spectral efficiency	Specialist skills	High
Manufacturing	Cisco (IE) + Fortinet OT	Catalyst IE, Cyber Vision; FortiGate Rugged	OT visibility	Industrial hardening	Cost	Medium-high
BFSI	Cisco + Palo Alto (or Check Point)	Catalyst/Nexus + PA-Series/Prisma	Compliance, references	Audit-ready	Highest cost	High
Healthcare	HPE Aruba / Juniper Mist	AP-700, ClearPass, Mist	IoT onboarding, NAC	Device profiling	Gateway licensing	Medium
Retail	Fortinet / Meraki	FortiGate + FortiAP; Meraki MX/MR	Thousands of small sites	Zero-touch provisioning	Subscription renewals	Low-Medium
Government	Cisco / HPE / Juniper (Make-in-India SKUs in India)	Catalyst; Aruba CX; EX/SRX	Certifications, trusted source	Compliance	Cost	High
Education	Extreme / HPE Aruba / Ruckus	Platform ONE; Aruba; RUCKUS One	Dense Wi-Fi, budgets	Pooled licences, eduroam	Smaller ecosystem (Extreme/Ruckus)	Low-Medium

19. Supply Chain & Commercial Considerations

Vendor	Lead time (typical 2026)	Local stock / spares	EoS / EoL policy & refresh cycle	Vulnerability response	Vendor / licensing lock-in	Cloud dependency
Cisco	2-8 wks campus; 8-16 wks DC/high-end	Excellent globally; strong India depots	EoS notice ~6 months; 5 yrs support after EoS; 5-7 yr refresh	PSIRT mature; frequent advisories	High (DNA subs, SGT, EA)	Medium (Meraki high)
HPE Aruba / Juniper	2-8 wks; DC 6-12 wks	Good; India manufacturing	Lifetime warranty on many switches; 5 yr support after EoS	Mature PSIRT/SIRT	Medium (Central/Mist)	Medium-High
Arista	4-12 wks; AI DC can be longer	Good in US/EU; thinner in India	Single EOS train; clear EoL	Strong	Low-Medium	Low (CVaaS optional)
Huawei	2-6 wks where available	Strong in Asia/MEA	Clear lifecycle	PSIRT; trust concerns	Medium	Medium
Fortinet	1-6 wks	Very good via distributors	EoO/EoS published; ~5 yr	Frequent critical CVEs; fast patches	Medium-High (Security Fabric)	Low-Medium
Extreme	2-8 wks	Moderate	Clear; pooled licences reduce stranded value	Good	Low (portable licences)	Medium
Palo Alto	2-6 wks	Good	5 yr	Strong	High	Medium (Prisma)
Ubiquiti	Often immediate (store) but stock-outs common	Retail channel	Short, informal	Community-driven	Low	Medium (UI account)
MikroTik	Immediate via distributors	Good	Long-lived RouterOS	Variable patch adoption	Low	Low

20. Risk Analysis

Higher score = higher risk (1 = very low, 10 = very high). Risk covers vendor lock-in, licensing changes, price increases, supply chain, geopolitics, regulation, security posture, cloud dependency, discontinuation, skills and partner dependency.

Vendor	Commercial risk	Technical risk	Principal risks
Cisco	6	3	Licensing changes/renewal uplift, EA true-forward, product renaming; technically mature.
HPE Aruba Networking	4	4	Platform convergence with Juniper Mist may change SKUs and UIs.
Juniper (HPE)	5	4	Portfolio rationalisation post-acquisition (e.g., overlapping campus/SD-WAN lines).
Arista Networks	4	3	Partner dependency for security; VeloCloud integration risk.
Huawei	9	4	Sanctions, NDAA §889, EU 5G Toolbox, India trusted-source rules; component supply risk.
Nokia	4	4	Enterprise channel thin; focus is SP.
Extreme Networks	5	4	Smaller scale; skills availability.
Fortinet	3	5	Recurring critical vulnerabilities in FortiOS/SSL-VPN require disciplined patching.
Palo Alto Networks	6	3	Subscription escalation; platformisation lock-in.
Check Point	4	4	Narrower roadmap outside security.
Versa Networks	5	4	Private-company scale; partner dependency.
VMware by Broadcom	9	5	Per-core VCF-only licensing, price increases, divestitures, partner changes.
Ciena	3	3	Specialist; low risk in its niche.
Dell Technologies	4	5	SONiC skills scarcity; networking is a small Dell business.
Ubiquiti	3	6	No enterprise TAC; tariff surcharges; cloud-account dependency.
Ruckus Networks	5	5	Ownership changes under CommScope restructuring.
MikroTik	4	7	Unpatched devices historically exploited; no TAC.
Cato Networks (added)	4	4	Single-cloud dependency; private company.
Zscaler (added)	4	4	Cloud dependency; outage blast radius.

21. Final Executive Comparison

Vendor	Tech	Security	SD-WAN	DC	WLAN	Autom.	Licensing	5Y TCO	India	Europe	US	Overall
Cisco	9	8	8	9	9	8	4	5	10	9	10	8.1
Fortinet	7	9	10	4	7	7	8	9	9	9	8	8.1
HPE Aruba Networking	8	7	8	6	10	8	6	7	8	9	8	7.8
Juniper (HPE)	8	7	7	9	9	10	6	6	7	8	8	7.8
Arista Networks	8	5	6	10	7	10	7	6	6	8	9	7.6
Huawei	8	7	7	8	9	7	7	9	4	4	1	7.6
Palo Alto Networks	7	10	8	2	1	8	4	4	8	8	9	7.3
Cato Networks (added)	5	9	9	1	1	7	7	7	6	8	8	7.2
Versa Networks	6	8	9	2	3	7	7	7	6	7	6	7.0
Extreme Networks	7	6	5	6	8	7	8	7	5	7	7	7.0
Check Point	6	9	5	2	1	6	5	5	7	8	7	6.4
Zscaler (added)	5	9	4	1	1	7	6	6	8	8	9	6.4
Nokia	7	5	4	8	3	9	6	6	6	8	6	6.3
Ruckus Networks	6	3	1	1	8	5	7	8	7	6	6	5.8
Ciena	6	3	2	5	1	8	6	6	6	7	7	5.8
VMware by Broadcom	6	6	2	7	1	7	2	3	5	5	5	5.7
Dell Technologies	5	3	1	7	2	6	7	8	6	6	6	5.4
Ubiquiti	4	3	2	1	6	3	10	10	6	6	7	5.2
MikroTik	4	3	2	1	2	4	10	10	6	6	4	4.7

Overall = weighted score from Section 14. Licensing and 5Y TCO: 10 = most favourable to the buyer. Regional scores reflect availability, partner depth, regulatory fit and support.

22. Final Ranking

22.1 Full-stack enterprise networking (campus + WLAN + WAN + DC)

7. **Cisco** — most complete and lowest-risk, highest cost.
8. **HPE Networking (Aruba + Juniper)** — closest functional parity, better AI/ops, ~15% lower TCO.
9. **Fortinet** — best value where security-led branch/campus dominates; weaker DC.
10. **Arista** — best DC; campus and SD-WAN credible but young.
11. **Extreme Networks** — simple licensing, education/healthcare strength.
12. **Huawei** — strong and cheap, but only where regulation and risk appetite allow.

22.2 Category leaders

Category	#1	#2	#3
Campus / WLAN	HPE (Aruba + Juniper Mist)	Cisco	Huawei (where permitted) / Extreme
Data center	Arista	Cisco	Juniper (Apstra)
SD-WAN	Fortinet	Cisco	HPE EdgeConnect / Palo Alto / Versa
SASE / SSE	Palo Alto Networks	Fortinet / Cato	Zscaler (SSE) / Netskope
NGFW	Palo Alto Networks	Fortinet	Check Point / Cisco
SP routing	Nokia	Juniper	Cisco
Optical	Ciena	Nokia (Infinera)	Cisco (Acacia)
AI/ops / automation	Juniper Mist	Arista CloudVision	HPE Aruba Central / Cisco
SMB / budget	Ubiquiti	Fortinet (small FortiGate)	MikroTik

23. Final Recommendation — 15 Questions Answered

#	Question	Answer	Rationale
1	Strongest overall Cisco alternative?	HPE Networking (Aruba + Juniper)	Only vendor matching Cisco across campus, WLAN, NAC, SD-WAN, SSE, routing, DC and firewall, with leading AIOps.
2	Best price/performance?	Fortinet (branch/campus/security); Arista (DC)	ASIC-based FortiGate bundles; Arista's throughput and reliability per dollar.
3	Lowest estimated 5-year TCO?	Fortinet among enterprise-grade stacks (Huawei comparable where permitted; Ubiquiti lowest for SMB only)	See Sections 12–13.
4	Strongest SD-WAN?	Fortinet (overall), with Cisco and HPE EdgeConnect close	Long-standing Gartner Leaders; Fortinet ranked #1 for secure branch modernisation in 2025 SASE Critical Capabilities (vendor-reported).
5	Strongest DC networking?	Arista	EOS, CloudVision, 400/800G AI fabrics.
6	Strongest WLAN/campus?	HPE (Aruba + Juniper Mist)	WLAN/NAC/AIOps depth; Cisco a close second.
7	Strongest security ecosystem?	Palo Alto Networks	NGFW + Prisma SASE + Cortex; Fortinet best integrated with networking.
8	Best automation/AIOps?	Juniper Mist (campus) / Arista CloudVision (DC)	Most mature AI-driven assurance and state-streaming automation.
9	Strongest in India?	Cisco	Installed base, partners, engineers; Fortinet and HPE Aruba strongest challengers.
10	Strongest in Europe?	Cisco, with HPE and Fortinet close	Partner depth; NIS2/GDPR-ready cloud regions.
11	Strongest in the US?	Cisco (enterprise), Arista (DC), Palo Alto (security)	Certifications and ecosystem.
12	Best for global enterprise?	Cisco (lowest risk) or HPE Networking (better TCO)	Global TAC/depots and consistent licensing.
13	Best for a telecom company?	Nokia (IP/optical) with Juniper/Cisco; Ciena for optical	SR/SRv6 scale and automation.
14	Best for cost-sensitive organisation?	Fortinet (enterprise-grade); Ubiquiti/MikroTik for SMB	Lowest TCO with acceptable support.
15	Recommendation for a new enterprise network today?	Best-of-breed by domain: HPE Networking or Cisco for campus/WLAN/NAC; Arista for DC; Fortinet or Palo Alto for security/SD-WAN/SASE	Standards-based (EVPN-VXLAN, OSPF/BGP, RADIUS) design keeps each domain replaceable; negotiate co-termed multi-year licences with renewal caps.

24. Procurement Recommendations

24.1 RFP structure

13. Issue a scenario-based RFP (like Scenarios A–D) and require a **5-year TCO** with every element priced separately: hardware, perpetual licences, term subscriptions, cloud management, security subscriptions, NAC, SASE, support, optics, services, training.
14. Require **list price, discount % and net price** per line, per region (INR/EUR/USD), excluding and including GST/VAT.
15. Demand **renewal price caps** (CPI or ≤5% per year) for all subscriptions and support for Years 2–7.
16. Require **co-termination** of all subscriptions and an annual true-up (not true-forward) mechanism.
17. Ask for **third-party optics support** or a fixed optics price list.
18. Evaluate with a weighted scoring model (Section 14) and a hands-on **proof of concept** (minimum 30 days).

24.2 Negotiation levers

- Competitive tension: at least three shortlisted vendors (e.g., Cisco, HPE, Fortinet) per domain.
- Quarter-end/fiscal-year-end timing (Cisco FY ends late July; HPE late October; Fortinet/Palo Alto vary).
- Competitive displacement and Make-in-India/PMA incentives (India).
- Bundle security + networking for platform discounts, but avoid committing beyond realistic device counts.
- Multi-year prepay only with price protection and termination-for-convenience on cloud services.

24.3 Regional procurement notes

Region	Key actions
India	Verify HSN classification and BCD with the distributor; prefer Make-in-India SKUs for government; confirm 4-hr depot coverage in Tier-2/3 cities; separate OEM support from SI AMC; budget 3–5% FX buffer.
Europe	Confirm EU data residency for cloud management and SASE logs; NIS2 supply-chain clauses; WEEE; country VAT treatment; local-language support.
United States	Check tariff surcharges on hardware; TAA/NDAA §889 compliance for public sector; state sales-tax exemptions; leverage reseller competition (CDW, SHI, Insight).

25. Top 10 Takeaways for a Network Architect / IT Procurement Manager

19. **The market has consolidated:** HPE now owns Juniper (July 2025) and Arista owns VeloCloud (July 2025); re-baseline any vendor shortlists written before mid-2025.
20. **Hardware is the smallest part of TCO** above branch size — subscriptions, support, SASE/NAC and labour dominate 5-year cost.
21. **Cisco remains the lowest-risk choice** but typically costs 25–45% more over five years than the cheapest enterprise-grade stack (and ~15–18% more than HPE).
22. **HPE Networking is the most complete Cisco alternative;** insist on roadmap commitments and price protection during the Aruba–Juniper integration.
23. **Arista is the default DC benchmark;** pair it with a security vendor because it has no first-party NGFW.
24. **Fortinet delivers the lowest TCO for secure branches and SD-WAN,** but run a disciplined patch programme for FortiOS edge devices.
25. **SASE is converging with SD-WAN;** single-vendor SASE (Palo Alto, Fortinet, Cato, Netskope, Cisco, HPE) simplifies operations — test PoP coverage in India and the EU.
26. **Huawei is technically strong and cheap, but regulatory exposure** (US NDAA, EU 5G Toolbox, India trusted source) makes it unsuitable for most multinational and regulated buyers.
27. **Negotiate renewals, not just the first purchase:** cap escalations, co-term everything, and avoid true-forward clauses.
28. **Design for portability:** standards-based protocols (OSPF/BGP, EVPN-VXLAN, RADIUS, gNMI) and automation (Terraform/Ansible) reduce lock-in and make future competition credible.

Appendix A — Sources (accessed 8 October 2026 unless dated otherwise)

Topic	Source	Date
HPE closes Juniper acquisition	Business Wire: businesswire.com/news/home/20250702141591/en; DatacenterDynamics; Nasdaq/Zacks	2 Jul 2025
Arista acquires VeloCloud	RCR Wireless (rcrwireless.com/20250702/business/arista-broadcom); The Register; SDxCentral; MarketScreener (completed 1 Jul 2025)	Jul 2025
Gartner WLAN MQ 2025	Huawei press release (e.huawei.com, 26 Jun 2025); Spotlight AR leaders list	25 Jun 2025
Gartner SD-WAN MQ 2024	HPE / Versa press releases (Business Wire, Oct 2024); Spotlight AR	30 Sep 2024
Gartner SASE Platforms MQ 2025	Palo Alto Networks, Fortinet, Cato Networks (PR Newswire), Netskope partner page	9 Jul 2025
Cisco Networking Subscription	cisco.com networking-subscription-ds.pdf; UK G-Cloud 15 Meraki pricing document (Jan 2026)	2026
Cisco Catalyst licensing (Network Ess./Adv., DNA/Catalyst term subs)	network-switch.com licensing guide; rfp.wiki Cisco Catalyst/Meraki comparison (verified Jun 2026); Redress Compliance Catalyst Center	2026
Cisco C9300-48P-E pricing	reach.dog / IT Bargain Center; wearesync.co.uk; eBay.de (Comstex)	Oct 2026 / Aug 2025
Meraki MX67 licences	CDW (LIC-MX67W-SEC-1YR); Hummingbird Networks (LIC-MX67-SEC-1YR)	Oct 2026
Fortinet FortiGate 60F bundle	SHI (FG-60F-BDL-950-36); datacenter360.ca; PB Tech	Oct 2026
Palo Alto PA-440 pricing	optinoc.com/blog/palo-alto-pa-440; Palo Alto LIVEcommunity (quote via reseller)	Oct 2026
Palo Alto subscription economics	atonementlicensing.com (enterprise pricing commentary)	2026
HPE Aruba Central pricing	SHI (JZ535AAE, JZ526AAE, S0V01AAE); CDW (S0T70AAE, JZ525AAE)	Oct 2026
Juniper Mist pricing	CDW (SUB-EX48-1S-5Y-COR etc.); Juniper Mist subscription documentation; Costbench (Jun 2026)	Oct 2026
Ubiquiti pricing	store.ui.com (USW-Pro-Max-48-PoE incl. surcharge, UI Care); SHI	Oct 2026
VMware/Broadcom licensing	Redress Compliance VCF/NSX guides; OTAVA; medhacloud	2026
Extreme Platform ONE licensing	Extreme Networks Switch Engine licensing guides (v33.x); ExtremeCloud IQ Universal Licensing	2025–2026
India customs/GST	KPMG Union Budget 2025-26 TMT POV; CESTAT order (Ingram Micro, Jan 2025); Lakshmikumaran & Sridharan (May 2025); eximpe HSN 8517; COAI statement	2025–2026
Huawei restrictions	Light Reading (India trusted source); EU 5G Toolbox progress report; Telecoms.com	2021–2025
Exchange rates	dollarrupee.in / 30rates (USD/INR ~94–96, Sept 2026); ValutaFX/CurrencyBeacon (USD/EUR ~0.86–0.87, Sept 2026)	Sep 2026

Appendix B — Disclaimer

This analysis is an independent market study produced on 8 October 2026 from public sources and professional judgement. It is not legal, tax, or investment advice. Vendor prices, licences and product availability change frequently; all 'Market estimate' figures are indicative ranges for budgeting and must be replaced by formal partner quotations. Gartner content is cited via vendor and press announcements; Gartner does not endorse any vendor.